Governance delay and the stability of harvested stocks: mobilising and protective feedback rules, and the review interval as a design parameter

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Abstract

Delays destabilise renewable-resource systems; those studied so far are ecological or informational. This paper analyses governance delay, the lag from observed decline to institutional response. A three-state model of harvested stock, filtered deficit memory, and bounded effort compares a mobilising rule, in which effort grows with effort already deployed, with a protective rule that tracks a quota by restoring harvest toward a cap.

For the mobilising rule, intermediate delay stabilises the equilibrium between two subcritical Hopf crossings, interval-certified near 3.7 and 150 yr. At the calibrated point, the protective rule is exponentially stable at every delay (a no-Hopf theorem). Under periodic review, annual protective review remains stable; its apparent 2.3-yr threshold is a discretisation artefact. Annual mobilising review is unstable and restabilises only above 6.5 yr through a Neimark–Sacker-type crossing. The five-regime attractor reading is numerical: two folds are certified only at the discrete-collocation level; the off-grid residual stage and continuous-delay lift remain open, so no rigorous saddle-node result for the delay equation is claimed. The basin-boundary transition leads toward an unverified large-amplitude attractor.

The model is formulated on a generic harvested stock; rule sign, deployment delay, and review cadence are design coordinates of this class, not transferable thresholds or empirical predictions. Northern cod's annual assessments, incremental response, 1992 moratorium, and 2024 reopening ground the timing scales, not coefficients. Within the model, institutional form and timing — not ecological lag alone — shape stability; more frequent review is not always safer, and the review interval is a local spectral design parameter.

Keywords: delay differential equations; Hopf bifurcation; institutional feedback; renewable resource management; fisheries governance; sample-and-hold control; regime shifts; maturation delay; recruitment dynamics

1. Introduction

1.1 Delays in the governance loop

Delayed feedback has a long history in resource dynamics. The classic results all place the delay inside the ecology itself:

- Hutchinson (1948) showed that a delayed self-limitation term destabilises logistic growth.
- Ezekiel (1938) documented the two-year lag between price signals and supply response in the pork cycle.

- Ludwig, Jones, and Holling (1978) placed delays and hysteresis at the centre of outbreak dynamics.
- Gurney, Blythe, and Nisbet (1980) gave the canonical stability analysis of the single-species delayed logistic equation.
- Costantino et al. (1995) demonstrated delayed-feedback cycles in controlled laboratory populations.
- A large contemporary literature continues this line, with delays placed in the ecological dynamics themselves: maturation and stage structure (Aiello and Freedman, 1990; Kuang, 1993), and harvested predator–prey systems whose paired ecological delays are Hopf objects in their own right (Zhang, Shen, and Chen, 2013).

The delay studied in this paper sits elsewhere: in the *institutional* loop — **governance delay**, the lag between the institution's observation of a decline and the response that finally acts on it. Stock assessment produces a signal of decline. The signal is filtered into a deficit memory. The memory drives a change in deployed extraction effort, with a deployment delay between decision and action. The behavioural basis for such lags is documented. Decision makers in renewable-resource experiments systematically misperceive the dynamics of the systems they govern, responding too slowly and in the wrong functional form (Moxnes, 1998). The modelling counterpart treats the delay as one of knowledge: delayed knowledge of the harvested population state destabilises bioeconomic equilibria and induces resource-harvester cycles, which harvester forecasting can dampen (Adamson and Hilker, 2020). Institutional design determines which signals reach which actors at which time (Ostrom, 1990). The management record supplies the field-scale counterpart: the northern cod collapse proceeded through years of assessment advice and incremental institutional response before the 1992 moratorium (Hutchings and Myers, 1994; Walters and Maguire, 1996), and the assessment-frequency literature now studies the review cadence itself as a managed design parameter with measured consequences for yield and depletion risk (Li, Bence, and Brenden, 2016; Peterson et al., 2022).

What the dynamical-systems literature has not supplied is an analysis of what this delay *does* to the stability of a harvested-stock system when the delay sits in the governance controller rather than in the ecology. Two questions follow, and the paper takes up both on a declared class of stock–memory–effort models. First, does the sign structure of the feedback law — mobilising versus protective — select between qualitatively different bifurcation behaviour? Second, what happens when the delay is replaced by periodic review (sample-and-hold)?

The institutional loop is not fisheries-specific, and neither is the analysed class. Assessment on a fixed cadence, a rule that converts the assessed signal into a quota or an effort decision, and a deployment lag between decision and action — this structure recurs across renewable-resource institutions, from quota-tracked harvests to capped withdrawals. It is the formal signature of a coupled human–natural system whose human component is a controller with memory. The model class of Section 2 is deliberately stock-agnostic: the state N is a renewable stock in material or biomass units, not a named fishery. Section 9 registers the same governance structure acting on a stock whose own ecology carries a maturation delay, so institutional and ecological delay are analysed within one frame. Fisheries supply the motivating instance — the setting in which periodic review is most literally institutionalised — and the management language of the two rules (effort escalation versus quota tracking) is the language of harvest control.

The overshoot mechanism analysed throughout has a one-sentence single-asset form. A hen produces a sustainable flow of eggs $S(N)$ each year; consumption D above $S(N)$ cannot be conjured, so the shortfall is covered by slaughtering the hen — the asset is destroyed to meet present demand. The loop is self-reinforcing. A deficit raises liquidation pressure, liquidation lowers the stock, the lower stock lowers sustainable yield, and the deficit grows. With the institutional deployment delay of this paper inserted into the loop, the deficit is observed, filtered, and answered with a lag. The hen's loop is then the delayed mobilising channel, whose

local bifurcation analysis occupies Sections 2–5; the protective channel is Section 6; sample-and-hold review, Section 7; the global numerics, Section 8. The delay does not create the shortfall; it decides when the response to it arrives. The phase-stabilised window of Section 5.1 is the interval of delays inside which the lag, not the feedback, holds the loop stable.

Not all liquidation channels are the same. Killing the hen removes a reproducing adult at once (culling the standing stock). Eating fertilised eggs suppresses next year's recruits without touching a living adult (recruitment suppression). Eating an apple often leaves the seed unharmed, so consumption and reproduction can be substantially decoupled. The first two channels are the two-channel mixture registered in Section 8.3 (M3-LC). That is why the channel assignment there requires age-, stage-, or replenishment-specific evidence rather than local spectra alone.

1.2 Contributions

1. **A gated three-state core.** A stock–memory–effort system (Section 2) in which a multiplicative saturation gate enforces the effort boundary exactly, making the admissible box forward invariant (Theorem 2.1). The model is a mathematical parameterisation, not a calibration to a named fishery; two reference parameter sets (Candidates A and B) serve as sensitivity anchors.

2. **The complete Hopf cubic and the even-pairs algebra.** The characteristic quasi-polynomial of the linearisation reduces to a cubic modulus condition with a phase relation (Theorem 4.1). At the interior equilibrium, a filter identity makes the cross term of the cubic vanish identically, so the total algebraic multiplicity of positive roots is even — generically zero or two distinct simple roots (Corollary 4.1). The two-frequency structure of the architecture is thereby an algebraic fact, not a numerical accident.

3. **Interval-certified crossings.** The two Hopf delays of the mobilising channel are certified by interval Newton iteration on the cubic with outward rounding — near 3.7 and 150 yr at Candidate A, with the 13-digit enclosures displayed in the table of Section 5.1. The certification tier is stated precisely: the enclosures certify the *local spectrum* of the cubic. The method of Church and Lessard (2022), which certifies full Hopf bifurcation for retarded functional differential equations, is the stronger standard and is not claimed here.

4. **The sign separation.** The mobilising law (gain growing with deployed effort, $C_Z > 0$) and the protective quota-tracking law (restoring gain, $C_Z < 0$) yield opposite local mathematics on the same stock–memory block. The mobilising channel carries the Hopf pair, both crossings subcritical (Section 5.2). The protective channel carries a **no-Hopf theorem** — all cubic coefficients positive, Descartes' rule and Routh–Hurwitz exclude every imaginary root, and the loop gain peaks at $0.08011 < 1$ — so exponential stability holds for every delay (Theorem 6.1). The two channels are interpolated, and a weighted small-gain theorem together with a conditional weight threshold show that delay-induced instability requires a sufficiently large mobilising weight (Section 5.4).

5. **The review interval as control.** Under sample-and-hold review with one explicit Euler step per period — the update timing fixed by the flow-then-update (pre-review/post-review) convention of Section 7 — the monodromy (the one-period return map of the reviewed loop) is computed in closed form (Proposition 6.2, Theorem 7.1). The protective channel is stable under annual review (spectral radius below one), and its apparent instability threshold near 2.3 yr is a discretisation artefact of the Euler update — provably not a Hopf of the continuous system (Sections 6.2 and 6.4). The mobilising channel is unstable under annual review (spectral radius just above one under both the Euler and the exact

update) and has its restabilising complex unit-circle crossing — the spectral signature of a Neimark–Sacker bifurcation, with nonlinear conditions not verified — at about 6.5 yr under the exact held-measurement update (Proposition 7.1). The Euler-reported half-century crossing and its companion -1 crossing are command-step artefacts. For a slow stock under this review operator, the review interval is a local spectral design parameter, and the continuous-delay recommendation does not transfer to periodic review (Section 7).

6. **Global numerics at declared status.** Continuation, Floquet-multiplier tracking, and basin tests give a five-regime attractor topology at declared certification tiers — and the tiers are stated precisely rather than as a single "certified" claim. Under the pre-registered five-regime campaign of Section 8.2, the lower boundary resolves into a single fold near 5.6 yr of one S-shaped branch — the subcritical-Hopf small arm and the stable large arm meet at one *discrete-collocation* Krawczyk-certified fold — and the upper boundary carries a second fold certified at the discrete-collocation level near 64.4 yr of a second S-branch. For both, the continuum off-grid residual stage and the continuous-delay lift remain **open**; a rigorous saddle-node result for the delay equation itself is not claimed (Beretka and Vas, 2020 covers specific classes; none is claimed for this system). The saddle-node-of-periodic-orbits classification is stated at its certification tier in Section 8.2. The stable returning arm of the upper branch is not generically reachable near the fold, so the capture onset near 149 yr is a basin boundary rather than a fold; the object captured there is a large-amplitude face cycle that no campaign could Fourier-collocate at $m = 64$ or 128, and the link from the stable returning arm's last record to that cycle over an $\approx$ 84-yr gap is an unverified conjecture. The five-regime reading of attractors (iv)–(v) is therefore a numerical reading that combines a certified-fold structure with an uncollocatable, unverified-capture region, and is reported as such.

1.3 Organization

Section 2 defines the model class and its admissible domain. Section 3 derives equilibria and the characteristic equation. Section 4 develops the Hopf cubic and the even-pairs algebra. Section 5 analyses the mobilising channel, and Section 6 the protective channel; together they carry the sign separation — opposite local mathematics on the same stock–memory block. Section 7 treats sample-and-hold review: the review interval is a spectral design parameter in its own right, and the reviewed loop is a hybrid system, not the delay equation sampled at the review interval. Section 8 reports the global numerics and their certification levels. Section 9 registers the maturation-delayed recruitment system — the applied analogue in which the same governance structure acts on a stock whose own delay is the maturation lag from spawning to recruitment — and records the companion stage-analysis fine-map bands and their nonlinear ground truth. Section 10 unifies the two channels in the loop-gain family. Section 11 discusses design consequences, the generality of the institutional-loop coordinates, the documented institutional timelines that ground the two timing coordinates, the ecological reading of the registered records, the relation to the early-warning literature, and open problems. Section 12 concludes.

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2. The Model Class

Notation. Throughout, N is a renewable stock (material or biomass units); Z is a filtered deficit-memory state (stock per time); and E is an extraction effort — a dimensionless institutional deployment intensity, not a conserved material or energy stock. The catchability q has units $(\text{effort}\cdot\text{yr})^{-1}$. The signal references Δ_{ref}, Z_{ref} and the offset δ have stock-per-time units; η and

δ_0 carry yr^{-1} and effort-per-time units. These assignments make each equation dimensionally homogeneous without treating effort as physical mass. The delay τ is a discrete action (deployment) delay. The filter timescale τ_m is an ordinary state relaxation time, not a second delay. The two sign channels are called the *mobilising channel* — the feedback law whose gain grows with deployment — and the *protective channel* — the quota-tracking law whose gain restores toward a cap. Periodic-review results use *sample-and-hold*: an Euler-reviewed zero-order-hold sampling of the delayed signal at intervals of length T_r. In plain words, the delayed signal is measured once at each review instant and held fixed between reviews, so the review interval T_r is the cadence at which the institution re-decides.

2.1 The gated three-state core

Let

$$S(N) = rN\left(1 - \frac{N}{K}\right), \qquad \mathrm{sp}_k(s) = \frac{1}{k}\log(1 + e^{ks}), \qquad \Phi_k(s) = \max\left\{0,\ \mathrm{sp}_k(s) - \frac{\log 2}{k} + \delta\right\},$$

where Φ_k is a shifted, non-negative signal map. The boundary-exact (gated) three-state core is

$$\begin{aligned} \dot{N} &= S(N) - qEN, \\ \dot{Z} &= \frac{1}{\tau_m}\big[\Phi_k(qEN - S(N)) - Z\big], \\ \dot{E} &= \left(1 - \frac{E}{E_{\max}}\right)\left[\eta E\left(\frac{Z(t-\tau)}{\Delta_{\mathrm{ref}}} - \frac{E}{E_{\max}}\right) + \delta_0 \frac{Z(t-\tau)}{Z_{\mathrm{ref}} + Z(t-\tau)}\right]. \end{aligned} \tag{1}$$

Three structural features of (1) carry everything that follows:

- **Depletion filtering.** The memory input is a smoothed version of the stock-decline rate: on this core the identity $qEN - S(N) = -\dot{N}$ holds exactly, so Φ_k filters $-\dot{N}$.
- **The multiplicative gate.** The gate $(1 - E/E_{\max})$ is essential to what follows. It is a hard saturation architecture, not a generic effort law, and it enforces $E \in [0, E_{\max}]$ by construction.
- **The institutional delay.** The effort law reads the filtered memory at one fixed lag — the deployment delay τ between decision and action.

The registered parameterisations are:

Parameter	Candidate A	Candidate B	Role
r	0.02	0.02	yr^{-1}; stock renewal
K	100	100	stock units; normalisation
q	0.001	0.001	$(\text{effort}\cdot\text{yr})^{-1}$
η	0.914	2.756	yr^{-1}; effort response
$E_{\max}$	30	26	effort units; saturation boundary
Δ_{ref}	1.0	1.0	stock yr^{-1}; signal scale
δ_0	0.01	0.01	effort yr^{-1}; baseline source
τ_m	5	5	yr; filter relaxation
Z_{ref}	1.0	1.0	stock yr^{-1}; signal scale
δ	$\log 2/10$	$\log 2/10$	stock yr^{-1}; signal offset
k	10	10	$(\text{stock yr}^{-1})^{-1}$; regularisation
τ	varied	varied	yr; action/deployment delay

These are mathematical parameterisations and sensitivity anchors, not a joint calibration to a named resource or institution. The institutional coefficients have not been independently identified from a field system. Candidates A and B are two points in the effort-response chart, not

rescalings of one class. For the reported pair $\delta = \log(2)/k$ the outer floor cancels algebraically ($\Phi_k(s) = \mathrm{sp}_k(s) > 0$ for every finite s), so the floor is inactive on every reported periodic orbit. This identity is parameter-specific; for $\delta \neq \log(2)/k$ floor contact must be checked orbit by orbit.

2.2 Admissibility

Theorem 2.1 (Forward invariance of the admissible region). *For the history class $\varphi \in C([-\tau, 0], \mathbb{R}^3)$ define*

$$\mathcal{D} = \{0 \le N \le K,\ Z \ge 0,\ 0 \le E \le E_{\max}\}.$$

Assume

(H1) the initial history lies in $\mathcal{D}$ — every history value, not only the endpoint.

Then every classical solution of (1) remains in $\mathcal{D}$ for as long as it exists.

Proof. Check the five boundary faces. (i) $N = 0$: $S(0) = 0$ and the harvest qEN vanishes, so $\dot{N} = 0$; the face is invariant. (ii) $N = K$: $S(K) = 0$, so $\dot{N} = -qEK \le 0$, inward. (iii) $Z = 0$: the source $\Phi_k(qEN - S(N)) \ge 0$ by construction of the non-negative signal map, so $\dot{Z} = \Phi_k(\cdot)/\tau_m \ge 0$, inward. (iv) $E = 0$: the gate factor is positive (away from $E = E_{\max}$) and the bracket is $\delta_0 Z_\tau/(Z_{\mathrm{ref}} + Z_\tau) \ge 0$, so $\dot{E} \ge 0$, inward. (v) $E = E_{\max}$: the gate vanishes, $\dot{E} = 0$; the face is invariant. The delay enters only through $Z(t - \tau)$, and the initial history lies in $\mathcal{D}$, so the delayed argument is non-negative throughout the first interval $[0, \tau]$. The vector field is therefore subtangential on the first interval by (i)–(v); the argument repeats by induction over $[n\tau, (n+1)\tau]$ by the method of steps (Hale and Verduyn Lunel, 1993). At $\tau = 0$ the same faces give subtangentiality of the ordinary differential equation directly. □

Corollary 2.1 (Boundedness and global continuation). *Let $\bar{Z} = \max\{\sup_{[-\tau,0]} Z,\ \Phi_k(qE_{\max}K)\}$. Then every solution satisfies $0 \le Z(t) \le \bar{Z}$; all three states remain in a bounded set on which the vector field is locally Lipschitz; and the solution continues for all $t \ge 0$.*

Proof. The Z-equation reads $\dot{Z} = d(\nu(t) - Z)$ with $d = 1/\tau_m$ and the monotone bounded input $\nu(t) = \Phi_k(qE(t)N(t) - S(N(t))) \le \Phi_k(qE_{\max}K)$. The letter ν is used only in this section, for the filter input; it denotes the Halanay decay rate in Section 10.2. Variation of constants gives $Z(t) = e^{-dt}Z(0) + d\int_0^t e^{-d(t-s)}\nu(s)\,ds \le e^{-dt}Z(0) + \bar{Z}(1 - e^{-dt})$, hence $Z(t) \le \max\{Z(0), \bar{Z}\} \le \bar{Z}$ by the definition of $\bar{Z}$. Nonnegativity of Z is part of Theorem 2.1. With $N \in [0, K]$, $E \in [0, E_{\max}]$, and $Z \in [0, \bar{Z}]$, the state lies in a compact set on which the right-hand side of (1) is locally Lipschitz: the logistic and the polynomial effort law are smooth; the signal map Φ_k is locally Lipschitz in general and smooth where its floor is inactive — in particular, at the registered relation $\delta = \log 2/k$ the floor cancels and $\Phi_k = \mathrm{sp}_k$ is C^1 everywhere. The solution therefore extends for all time. Local Lipschitz regularity suffices for the existence, uniqueness, and continuation arguments; the C^1 statements are needed only where bifurcation coefficients are computed, and there the floor-inactive parameter sets of Section 5.2 are the declared scope. One further observation is recorded here because Section 3 uses it. On the invariant extinction face $N = 0$ the memory input is $\Phi_k(0) = \delta$, so Z relaxes to δ, and the extinction rest carries the same admissible positive effort root as the interior branch: institutional memory sustains baseline commanded effort against zero realised harvest. Effort is an institutional deployment intensity, not a conserved stock; the core makes no closed-effort-energetics claim, and a materially closed application must add and donor-limit that support explicitly. □

2.3 The registered model family

The reference member of the family is **M3-B (boundary-exact gated)** — the core (1) itself, whose multiplicative saturation gate enforces $E \in [0, E_{\max}]$ exactly (Theorem 2.1). Its crossings,

folds, and basins are the paper's reference records (Sections 5.1 and 8.3). Four further variants delimit the robustness of those results. No invariant set, equilibrium formula, local threshold, periodic branch, or admissibility result transfers between them without a separate argument.

- **M3-U (ungated).** The same ecology with the effort law without the outer gate; $E_{\max}$ is then a self-limitation scale only. The gate factor is the algebraic source of threshold relocation between the variants (Section 3.2).
- **M3-LC (two-channel stock law).** The logistic renewal is written as a gross birth–mortality decomposition $B(N) = S(N) + \kappa r N$, $M(N) = \kappa r N$ ($\kappa \geq 0$), and the pressure qEN is split by $\psi \in [0, 1]$:

$$\dot{N} = \max\{0,\ B(N) - (1-\psi)qEN\} - M(N) - \psi qEN. \tag{2}$$

Here ψqEN is a realised removal of standing stock and $(1-\psi)qEN$ is demographic suppression — a prevented inflow, not a material transfer out of the adult-stock compartment. Equation (2) is a phenomenological stock equation, not a closed mass ledger. Whenever the recruitment floor is inactive, (2) reduces exactly to $\dot{N} = S(N) - qEN$. The floor never binds at the interior equilibrium (the binding condition reduces to $-\psi S(N) > \kappa r N$, impossible), so the equilibrium, the Jacobian, the characteristic equation, and both Hopf points are independent of ψ and κ. Local equality does not imply excursion equality: the floor truncates recruitment on large excursions, and the two channels diverge there (Section 8).

- **M4-A (active-pool extension).** Logistic renewal is replaced by $R(N, A) = rN(1 - N/K)\, A/(A + A_0)$ with a dynamic active-support pool $\dot{A} = -B(N, A) + \omega_A(A^{\mathrm{eq}} - A)$ and a fully declared donor-limited gross draw B. The relaxation term makes this a reduced open-pool model unless its donor/receiver reservoir is included explicitly. Freezing A is not a justified fast-variable elimination at the baseline $\omega_A = 10^{-3}$ yr^{-1}.
- **MPF (primitive-flux core).** Living biomass X, detritus U, active material $A = \mathcal{M} - X - U$, primitive fluxes $g(X, A) = \mu X A/(K_A + A)$, $m(X) = dX + cX^2$, $h(X, E) = qEX$, signed memory $\dot{Z} = (-\dot{X} - Z)/\tau_m$, the detritus equation $\dot{U} = m(X) - \gamma_U U$ — the detritus compartment receives the full mortality flux $m(X) = dX + cX^2$ and exports it at the specific rate $\gamma_U > 0$, yr^{-1}; this is the equation the invariance argument below uses — and the bounded effort law $\dot{E} = (1 - E/E_{\max})[\eta E Z(t-\tau)/\Delta_{\mathrm{ref}} + \delta_0 - \eta E^2/E_{\max}]$. The core has a signed zero-equilibrium memory ($Z^* = 0$, a true constant baseline), detritus, and a different effort equilibrium; it is not (1). On the boundary $X + U = \mathcal{M}$ the active pool is $A = 0$ and $g(X, 0) = 0$, so $\frac{d}{dt}(X + U) = -qEX - \gamma_U U \leq 0$, which — with $g(0, A) = 0$, $m(0) = 0$, and the donor-limited flux assumptions — proves forward invariance of the simplex $\{X \geq 0,\ U \geq 0,\ X + U \leq \mathcal{M}\}$ for the ecological subsystem under admissible effort.

2.4 The four-state working core and its relation to (1)

The working four-state core restores the active abiotic pool as a state: $\dot{N} = R(N, A) - qEN$, $\dot{A} = -B(N, A) + \omega_A(A^{\mathrm{eq},W} - A)$ with $A^{\mathrm{eq},W} = A^{\mathrm{eq,intrinsic}} + \kappa_A K/\omega_A$, and the Z and E equations of (1) unchanged except that $S(N)$ is replaced by $R(N, A)$. At the baseline $\omega_A = 10^{-3}$ yr^{-1}, $\kappa_A = 0.05$ yr^{-1}, $A_0 = 0.01K$, $A^{\mathrm{eq,intrinsic}} = 0.5K$, so $A^{\mathrm{eq},W} = 5050$.

Two closure statements fix its status. First, in the ideal large-reservoir limit $\sigma_{\mathrm{geo}} = 1$ — where $\sigma_{\mathrm{geo}} \in (0, 1]$ is the geological-reservoir adequacy ratio, a dimensionless companion-ledger quantity measuring how far the geological reservoir sustains the working core's donor draw ($\sigma_{\mathrm{geo}} = 1$ the ideal limit, $1 - \sigma_{\mathrm{geo}}$ the scale of the finite-reservoir perturbation) — the specialised system satisfies the working-core equations exactly (an algebraic identity check),

with detritus a driven auxiliary that does not feed back; for finite geological reservoirs the vector field is perturbed by $O(1-\sigma_{\text{geo}})$ and U feeds back. Second, detritus slaving under a fixed intrinsic target is an $O(\varepsilon_U)$ singular-perturbation estimate on compact intervals — a standard Tikhonov/Fenichel-type argument (Hale and Verduyn Lunel, 1993, Ch. 9, Diekmann et al., 1995); ε_U is the fast-variable slaving parameter, of the order of the slow-to-fast timescale ratio r/γ_U, with γ_U the detritus export rate of the MPF core, Section 2.3. At the baseline $\gamma_U/r = 10$ detritus is an order of magnitude faster than the stock, so the fast-variable slaving parameter $r/\gamma_U \approx 0.1$ is small and the compact-interval estimate holds. It is only moderately small, however, so it does not control global periodic orbits.

At Candidate A the quasi-steady (QSS) core has a positive low-A equilibrium $(N^*, A^*) \approx (23.85, 0.159)$ and no high-A near-logistic exploited equilibrium (the high-stock branch gives $A^* \approx -137$, inadmissible). The QSS core is a distinct singular-limit object: a valid limit that is not dynamically connected to the high-A working equilibrium used for the reported thresholds, and the two objects are never merged. The working core is declared as an open projection: omitted turnover is routed to a diagnostic detritus/inert sink; the reduced (N, A, Z, E) trajectory is not mass-closed by itself; and its global periodic results are model-version-specific. The working equilibrium is a frozen-donor quasi-equilibrium sustained by geological support of order $\omega_A(A^{\text{eq},W} - A^*) \approx 4.652$ stock units yr^{-1}, whose cumulative donor change is $4.652\,T/A^{\text{geo}}$ — 1.2% on a century horizon at the lower geological ratio $A^{\text{geo}}/A^* = 10^2$ and 0.12% at 10^3.

Remark (companion-interface numbers). *The quantities $A^{\text{eq},W} = 5050$, the geological support ≈ 4.652 stock units yr^{-1}, and the century draws 1.2%/0.12% are interface values carried from the companion material-ledger paper — where the closed mass ledger and its quasi-steady slaving target are the subjects — for cross-paper consistency. They are not hypotheses of any theorem of this paper, and they do not make the working-core quasi-equilibrium a rest point of a closed ledger: the four-state working core is an open projection, sustained by continuing geological support, precisely as stated above. If this paper is read alone, the four-state core should be taken as an open working model, with the 4.652 accounting a remark citing the companion rather than a structural result.*

The finite-time connection between the four-state and three-state cores is the following bound. It justifies using (1) for local, near-equilibrium questions on institutional timescales — but not the large-amplitude cycle or its period, and not spectral quantities. A trajectory bound does not transfer to Hopf thresholds, so the proximity of the four-state crossings to the three-state values (with τ_- 3.2% higher and τ_+ 0.2% lower than the three-state values, Section 8.4) is reported as a numerical observation, not derived from this bound.

Lemma 2.1 (Frozen-active-pool approximation). *Let (N, Z, E) solve (1) and let $(\tilde{N}, \tilde{Z}, \tilde{E})$ solve the four-state core from the same initial data, with $A(t) \geq A_{\min} > 0$ on $[0, T]$. Measure state differences in the scaled norm*

$$\|\xi\|_* = \frac{|\Delta N|}{K} + \frac{|\Delta Z|}{rK} + \frac{|\Delta E|}{E_{\max}},$$

where $\Delta N = \tilde{N} - N$, $\Delta Z = \tilde{Z} - Z$, $\Delta E = \tilde{E} - E$ (stock, flux, and effort scales respectively). Assume

(H1) the vector field of (1) is Lipschitz on the admissible region.

Then

$$\sup_{t\in[0,T]} \|\xi(t)\|_* \leq C_T \frac{A_0}{A_{\min} + A_0}$$

with C_T a constant depending on the structural parameters and T. This is an inner approximation on $[0, T]$, not a Tikhonov reduction.

Proof. The four-state renewal differs from the saturated renewal by

$$R(N, A(t)) - S(N) = rN\left(1 - \frac{N}{K}\right)\left(\frac{A(t)}{A(t) + A_0} - 1\right) = -rN\left(1 - \frac{N}{K}\right)\frac{A_0}{A(t) + A_0},$$

so the mismatch is bounded pointwise by $\frac{rK}{4} \cdot \frac{A_0}{A_{\min}+A_0}$ (the renewal factor is at most $rK/4$ and the saturation deficit is at most $A_0/(A_{\min} + A_0)$). The Z and E equations of the two cores coincide in form, so their difference equations are driven only by ΔN and by the renewal mismatch $\sigma(t) = R(\tilde{N}, A(t)) - S(\tilde{N})$, with $|\sigma(t)| \le \frac{rK}{4}\frac{A_0}{A_{\min}+A_0}$. The difference system is retarded (the E-row contains $\Delta Z(t-\tau)$), so the estimate is taken in the history norm: with $Y(t) = \sup_{s\in[-\tau,t]} \|\xi(s)\|_*$, Lipschitz continuity of the common vector field gives the retarded differential inequality

$$Y(t) \;\le\; C_1 \int_0^t \left(Y(s) + \frac{A_0}{A_{\min} + A_0}\right) ds,$$

for the structural constant C_1 — the scaled-norm Lipschitz constant of the difference vector field, finite by the Lipschitz hypothesis of the statement. The retarded Gronwall estimate (Hale and Verduyn Lunel, 1993, Ch. 6) then yields $\sup_{s\in[-\tau,T]} \|\xi(s)\|_* \le C_T A_0/(A_{\min} + A_0)$ on $[0, T]$, with $C_T = C_1 T e^{C_1 T}$ after absorbing constants; identical initial data give $\xi(0) = 0$. □

2.5 The protective channel and the two-channel interpolation

A protective institution is the quota-tracking law

$$\dot{E} = \left(1 - \frac{E}{E_{\max}}\right)\eta_p\big(E_{\text{cap}}(Z(t - \tau_p)) - E\big), \tag{3}$$

where E_{cap} is C^2, positive, strictly decreasing, with the calibration $E_{\text{cap}}(Z) = E_0 Z_{\text{ref}}/(Z_{\text{ref}} + Z)$ and $E_0 = E_A^*(Z_{\text{ref}} + \delta)/Z_{\text{ref}}$. The calibration places the unique interior rest of (3) on the stock–memory block of (1) at the Candidate A point $(N^*, Z^*, E^*) = (89.55188, \delta, 2.08962)$, so the stock–memory block is identical to (1)'s and only the effort law changes. At that rest $E_{\text{cap}}(\delta) = E^*$ and the linear gains are

$$C_E = -\left(1 - \frac{E^*}{E_{\max}}\right)\eta_p, \qquad C_Z = \left(1 - \frac{E^*}{E_{\max}}\right)\eta_p E'_{\text{cap}}(\delta),$$

which at $\eta_p = \eta_A = 0.914$ give $C_E = -0.850336$ and $C_Z = -1.661702$. Both signs are those of a restoring quota, not of scarcity mobilisation; the mobilising counterpart of (1) at the same point has $C_Z = +1.785$ and $C_E = -0.0595$ (Section 3.2). The channel-separation object of this paper is exactly this sign discipline.

The two-channel interpolation replaces the effort law by

$$\dot{E} = \left(1 - \frac{E}{E_{\max}}\right)\left[\chi_m F_m(E, Z(t - \tau_M)) + \chi_p \eta_p\big(E_{\text{cap}}(Z(t - \tau_p)) - E\big)\right], \tag{4}$$

with $\chi_m, \chi_p \ge 0$; the pure mobilising channel is $(\chi_m, \chi_p) = (1, 0)$, pure protection $(0, 1)$. The mobilising bracket is

$$F_m(E, Z) = \eta E\left(\frac{Z}{\Delta_{\text{ref}}} - \frac{E}{E_{\max}}\right) + \delta_0 \frac{Z}{Z_{\text{ref}} + Z},$$

the bracket of the mobilising law (1) written with its Z-argument displayed; τ_M is the mobilising deployment delay and τ_p the protective one. Throughout, the memory-filter timescale keeps its own symbol τ_m (used in the filter $\dot{Z} = (\Phi_k(\cdot) - Z)/\tau_m$); the two are distinct objects, distinguished by subscript wherever they co-occur.

3. Equilibria and the Characteristic Equation

3.1 The interior equilibrium and the extinction face

At any interior equilibrium $qE^*N^* = S(N^*)$, the signal argument vanishes, the floor is inactive, and

$$Z^* = \Phi_k(0) = \delta,$$

independent of τ_m, k, and (N^*, E^*). Substituting into $\dot{E} = 0$ (away from $E = E_{\max}$) produces

$$-\frac{\eta}{E_{\max}}(E^*)^2 + \eta\frac{\delta}{\Delta_{\text{ref}}}E^* + \delta_0\frac{\delta}{Z_{\text{ref}} + \delta} = 0,$$

whose constant term is positive and quadratic coefficient negative, so there is exactly one positive root. Admissibility requires $0 < E^* < \min\{E_{\max}, r/q\}$, and then $N^* = K(1 - qE^*/r)$, positive if and only if $qE^* < r$. The equilibrium is independent of τ and k, which does not make its stability delay-independent. One certified value is used throughout: at Candidate A the positive quadratic root is $E^* = 2.08962$ and $N^* = K(1 - qE^*/r) = 89.55188$, with $Z^* = \delta \approx 0.0693$. (The four-state donor-limited equilibrium of Section 8.4, $N^* = 89.52562$, $A^* = 397.8665$, is a distinct object and is never used as the three-state rest.)

On the extinction face $N = 0$ the same zero raw signal gives $Z = \delta$, and the gated law has both the interior-effort extinction rest $(0, \delta, E^*)$ — when the interior root is admissible — and the boundary rest $(0, \delta, E_{\max})$ created by the multiplicative gate. The stock-direction eigenvalue at either branch is $r - qE$. The interior positive-stock branch exchanges stock-direction stability with the interior-effort extinction branch at the transcritical point $r = qE^*$, while the $E = E_{\max}$ boundary branch is classified separately and is not part of that exchange. The survival condition $r > qE^*$ is identical to $N^* > 0$; interior and extinction branches are not simultaneously stable.

3.2 The characteristic quasi-polynomial

With $x = N - N^*$, $z = Z - Z^*$, $e = E - E^*$, $\text{sp}'_k(0) = 1/2$, and the floor inactive at $\delta > 0$, the linearisation of (1) is

$$\dot{x} = A_N x + A_E e, \qquad \dot{z} = B_N x + B_E e - \frac{1}{\tau_m} z, \qquad \dot{e} = C_E e + C_Z z(t - \tau),$$

with

$$A_N = r\left(1 - \frac{2N^*}{K}\right) - qE^*, \qquad A_E = -qN^*,$$
$$B_N = \frac{qE^* - S'(N^*)}{2\tau_m}, \qquad B_E = \frac{qN^*}{2\tau_m},$$
$$C_E = \left(1 - \frac{E^*}{E_{\max}}\right)\eta\left(\frac{\delta}{\Delta_{\text{ref}}} - \frac{2E^*}{E_{\max}}\right), \quad C_Z = \left(1 - \frac{E^*}{E_{\max}}\right)\left[\frac{\eta E^*}{\Delta_{\text{ref}}} + \frac{\delta_0 Z_{\text{ref}}}{(Z_{\text{ref}} + \delta)^2}\right].$$

The gate factors $(1 - E^*/E_{\max})$ distinguish the gated from the ungated variant and are the algebraic source of threshold relocation between the two. Substituting the modal ansatz $(x, z, e)e^{\lambda t}$ and expanding the 3×3 characteristic determinant along the third row gives the characteristic quasi-polynomial

$$P(\lambda) - C_Z L(\lambda) e^{-\lambda\tau} = 0, \qquad P(\lambda) = (\lambda - A_N)(\lambda + d)(\lambda - C_E), \qquad L(\lambda) = B_E(\lambda - A_N) + A_E B_N, \tag{5}$$

with $d = 1/\tau_m$. At the interior equilibrium the filter identities hold:

$$B_N = -\frac{A_N}{2\tau_m}, \qquad B_E = -\frac{A_E}{2\tau_m},$$

because the deficit signal is $qEN - S(N) = -A_N x - A_E e + o(\cdot)$ and $\mathrm{sp}_k'(0) = 1/2$. Consequently

$$L(\lambda) = B_E(\lambda - A_N) + A_E B_N = B_E\lambda - B_E A_N - A_E\frac{A_N}{2\tau_m} = B_E\lambda,$$

since $B_E A_N = A_E B_N$. This cancellation — the even-pairs algebra below — is the structural identity of the architecture.

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4. The Complete Hopf Cubic

The Hopf-frequency equation of the linearised system (5) takes a particularly clean form on this architecture: a cubic in $x = \omega^2$ for the modulus, plus a phase relation in τ. The cubic has at most three positive roots, so the system supports at most three Hopf-frequency families. The interior equilibrium's filter identity then collapses one structural term, and the resulting even-pairs algebra explains why, generically, two crossings are observed rather than one or three.

Theorem 4.1 (Cubic modulus condition and phase branches). *Consider the linearisation (5). Assume*

(H1) $C_Z L(i\omega) \neq 0$ at every candidate imaginary root. This is automatic on the declared family: by the filter identity of Section 3.2, $L(i\omega) = B_E i\omega$, so $C_Z L(i\omega) = 0$ only at $\omega = 0$, which is excluded by $\omega > 0$; the degenerate case $C_Z B_E = 0$ is excluded by the standing assumptions. Hypothesis (H1) is stated for completeness, to make the phase balance (7) well defined; it imposes no additional restriction here.

Then $\lambda = i\omega$ ($\omega > 0$) is a characteristic root of (5) if and only if $x = \omega^2$ is a positive root of

$$H(x) = (x + A_N^2)(x + d^2)(x + C_E^2) - C_Z^2\left[B_E^2 x + (A_E B_N - A_N B_E)^2\right] = 0, \tag{6}$$

and

$$\tau = \frac{-\arg\{P(i\omega)/(C_Z L(i\omega))\} + 2\pi k}{\omega} > 0, \qquad k \in \mathbb{Z}. \tag{7}$$

A cubic has at most three positive roots, so there are at most three Hopf-frequency families; higher branches recur within a family as $\tau_{n,0} + 2\pi k/\omega_n$ and are not additional frequencies. The certified branch pairs are separated — $\tau_{1,k} \neq \tau_{2,\ell}$ over the declared search range — so simultaneous double-Hopf collisions are excluded on the computed set (computed on the declared search range; not an algebraic consequence of the cubic). Simplicity and transversality are verified separately for each reported crossing.

Proof. At $\lambda = i\omega$, equation (5) reads $P(i\omega) = C_Z L(i\omega) e^{-i\omega\tau}$, so the moduli must match:

$$|P(i\omega)|^2 = C_Z^2 |L(i\omega)|^2. \tag{8}$$

Compute $|P(i\omega)|^2 = |(i\omega - A_N)(i\omega + d)(i\omega - C_E)|^2 = (\omega^2 + A_N^2)(\omega^2 + d^2)(\omega^2 + C_E^2)$, and $L(i\omega) = B_E i\omega + (A_E B_N - A_N B_E)$, whence $|L(i\omega)|^2 = B_E^2\omega^2 + (A_E B_N - A_N B_E)^2$. Substituting into (8) and writing $x = \omega^2$ gives exactly (6). Given a positive root x of (6), the phase condition reads $e^{-i\omega\tau} = P(i\omega)/(C_Z L(i\omega))$; since both sides have unit modulus on the root locus, τ is recovered as (7) up to the $2\pi k$ ambiguity. A cubic has at most three positive roots, giving at most three frequency families; for a fixed family ω_n the delays $\tau_{n,0} + 2\pi k/\omega_n$ recur with period $2\pi/\omega_n$ in τ, and are branches of the same frequency, not additional frequencies. Candidates

qualify as Hopf crossings only after simplicity and transversality are verified separately; the cubic determines neither criticality nor global folds. □

Corollary 4.1 (Even pairs). *At the interior equilibrium of (1), the cross term vanishes identically, $A_E B_N - A_N B_E \equiv 0$, so*

$$H(x) = (x + A_N^2)(x + d^2)(x + C_E^2) - C_Z^2 B_E^2 x.$$

Assume $A_N d C_E \neq 0$ (nonzero filter rate, memory rate, and equilibrium effort derivative). Then $H(0) = A_N^2 d^2 C_E^2 > 0$ and $H(x) \to +\infty$; the total algebraic multiplicity of the strictly positive roots of H is even, and generically H has zero or two distinct simple positive roots. The double-root boundary — the birth or annihilation of a Hopf-frequency pair — solves $H(x) = H'(x) = 0$.

Proof. The identities $B_N = -A_N/(2\tau_m)$, $B_E = -A_E/(2\tau_m)$ give $A_E B_N - A_N B_E = -A_E A_N/(2\tau_m) + A_N A_E/(2\tau_m) = 0$. With the cross term zero, H has the stated form, and $H(0) > 0$ follows from the nondegeneracy assumption. A polynomial with positive value at 0 and positive leading coefficient crosses the positive axis an even number of times counting multiplicity, which is the multiplicity claim; a single positive double root is the degenerate residual case and sits on the boundary $H = H' = 0$. □

For both Candidate A and Candidate B the cubic has exactly two positive roots on both the gated and the ungated variant; the certified local spectrum is stated in Section 5.1.

4.1 The scalar archetype

Two general statements frame the delay mathematics. First, the scalar Hayes result (Hayes, 1950; Hale and Verduyn Lunel, 1993, Ch. 3): for $\dot{x} = -ax - Bx(t-\tau)$ with $a > 0$, $B \geq 0$, the zero equilibrium is stable for all $\tau \geq 0$ if $B \leq a$; if $B > a$ it is stable for $\tau < \tau_{\text{crit}} = \arccos(-a/B)/\sqrt{B^2 - a^2}$ and unstable beyond, with *no restabilisation* as τ grows — the crossing is destabilising. This is the scalar mechanism by which loop delay destroys stability that undelayed feedback sustains. Second, no delay conclusion is sign-free: the systems $\dot{x} = -ax - bx(t-\tau)$ and $\dot{x} = -ax + bx(t-\tau)$ $(a, b > 0)$ have different feedback signs and different stability properties. The named systems of this paper instantiate the two signs — the mobilising law of (1) carries $C_Z > 0$, the protective law (3) carries $C_Z < 0$ — and Sections 5–6 show that the local mathematics separates accordingly.

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5. The Mobilising Channel

The mobilising channel is the effort law of (1) with positive delayed gain on the deficit memory. Its behavioural reading is direct: when an institution sees a deficit signal, it deploys more extraction effort — a fleet expansion, a subsidy release, an open-access licence issuance — and the deployment is realised only after a delay. The mathematics of this section says two things. First, the undelayed gated version of such a law is already unstable. Second, an interval of intermediate delays — rather than the short-delay regime familiar from the scalar Hayes example — is the window in which the loop is stabilised by phase. The implication for governance is direct: on this architecture the institutional deployment delay is a phase filter that can either stabilise or destabilise depending on its value. It is not a one-directional hazard.

5.1 Local crossings and interval-certified delays

Two terms are used throughout, and both are defined here for the broad reader. A *Hopf crossing* is a parameter value at which a complex-conjugate pair of characteristic roots crosses the imaginary axis and oscillatory dynamics are born. *Interval-certified* means the crossing delay

is enclosed by interval Newton iteration with outward rounding, so the enclosure is guaranteed to contain the exact root of the registered equations (Moore, 1979; Cloud, Moore, and Kearfott, 2009).

The complete cubic search (Theorem 4.1) with separately verified simplicity and transversality, independently checked by direct root tracking of the quasi-polynomial, gives the crossing pairs:

System	Candidate A τ_- / τ_+ (yr)	Candidate B τ_- / τ_+ (yr)
M3-U (ungated)	6.8814 / 132.3749	6.2136 / 76.2906
M3-B (gated)	3.67 / 150.36	5.5128 / 80.4245
M3-B (gated), interval-certified	$\tau_- \in [3.6661490142739, 3.6661490142743]$, $\tau_+ \in [150.3584773101408, 150.3584773101414]$	gated: $\tau_- \in [5.5128407314433, 5.5128407314436]$, $\tau_+ \in [80.4245267142270, 80.4245267142276]$; ungated: $\tau_- \in [6.2135987340180, 6.2135987340183]$, $\tau_+ \in [76.2906356879512, 76.2906356879518]$

The interval certificates are interval-Newton enclosures of the simple positive roots of H in $x = \omega^2$ (width $\leq 4 \times 10^{-17}$ in x) followed by branch-safe interval evaluation of the phase relation (7). The delay is the interval evaluation of the phase formula at a certified positive root of H, not a root of an argument formula. The registered interval pipeline uses outward rounding and interval transcendentals, checks simplicity and transversality signs — the lower crossing stabilising, $\mathrm{d}\,\mathrm{Re}\,\lambda/\mathrm{d}\tau < 0$; the upper crossing destabilising — and reproduces the displayed Candidate A intervals exactly on re-execution.

The certification tier is scoped precisely. These are certified enclosures of the *local spectrum* of the cubic — a strictly weaker statement than a certified Hopf bifurcation for retarded functional differential equations in the sense of Church and Lessard (2022) and Church and Queirolo (2024), whose radii-polynomial methods certify the bifurcation itself. No such full certificate is claimed here, and none is claimed for any global fold (Section 8, and the supplementary material). The interval machinery itself is standard (Moore, 1979; Cloud, Moore, and Kearfott, 2009). The pair is registered as gate G5 of the registration campaign — the gate log is deposited with the supplementary material, S9. The registered compute-core Hopf pair 3.666149 / 150.358477 yr — the base core (1)'s institutional-delay certificates, reproduced by the recovered compute core (Supplementary S9.5) — certifies this pair; it is a record of the compute core, not an object of the delayed-recruitment system of Section 9.

The undelayed gated mobilising law is already unstable ($\mathrm{Re}\,\lambda > 0$ at $\tau = 0$): the undelayed characteristic polynomial $P(\lambda) - C_Z B_E \lambda = (\lambda - A_N)(\lambda + d)(\lambda - C_E) - C_Z B_E \lambda$ in the symbols of Section 3.2 — the $-C_Z B_E \lambda$ term being exactly what the filter identity leaves of the delay coupling at $\tau = 0$ — at the declared Candidate A constants violates the Routh–Hurwitz condition. The linear coefficient is reduced by $C_Z B_E$, so the product of the first two coefficients falls below the constant term, which is the algebraic signature of the shortest-delay instability; and $P(\lambda) - C_Z B_E \lambda = 0$ is the undelayed case of (5). Institutional delay acts as a phase filter that opens the phase-stabilised window (τ_-, τ_+) and closes it again at τ_+. Delay-amplified instability refers to the upper crossing and the bistable windows, not to delay creating the short-delay instability. Enforcing the effort boundary relocates the local thresholds by approximately 47% (lower) and 14% (upper) at Candidate A without changing the equilibrium — thresholds do not transport between the gated and ungated variants, and the gate's threshold relocation is a registered comparison, not a calibration.

5.2 Lyapunov coefficients and criticality

The first Lyapunov coefficient at a Hopf point of the gated three-state core is the Hassard–Faria–Magalhães cubic (Hassard, Kazarinoff, and Wan, 1981; Faria and Magalhães, 1995) evaluated from the exact second and third derivatives of the vector field at equilibrium, under unit Hermitian normalisation of the right eigenvector v and its adjoint w, with the normalisation $w^*\Delta'(i\omega)v = 1$ (w^* the conjugate transpose; the eigenvector pair is denoted v, w so that q remains the catchability throughout):

$$\ell_1(\tau_-^{\mathrm{A}}) = +5.75 \times 10^{-5}, \qquad \ell_1(\tau_+^{\mathrm{A}}) = +3.55 \times 10^{-4},$$

both subcritical at gated Candidate A; the ungated Candidate B lower crossing is supercritical, $\ell_1(\tau_-^{\mathrm{B}}) = -9.84 \times 10^{-5}$ — hence no lower fold for that class — with $\ell_1(\tau_+^{\mathrm{B}}) = +2.19 \times 10^{-3}$. The subcritical small branch satisfies $\|N - N^*\| \sim C\sqrt{\tau - \tau_-}$ (slope 29.8 in amplitude-squared, $R^2 = 0.994$; the collocated orbit at $\tau = 3.700$ has residual $\sim 10^{-7}$ and escapes onto the large cycle by roundoff alone). These are numerical evaluations of the coefficient formulas at declared parameter points — computational results, stated as such.

Two status distinctions matter here. First, within the registered model family the criticality statements obtained from branch scaling — amplitude exponent 0.47 and surrogate cubic coefficient $\approx 3.9 \times 10^{-6}$ near the lower gated crossing — are inferred numerical classifications, not first Lyapunov coefficients from a centre-manifold calculation; the ℓ_1 values above are the computed coefficients for the gated core. Second, criticality is not invariant under the regularisation: the first Lyapunov coefficient contains $\mathrm{sp}_k''(0) = k/4$, so the reported $\ell_1(\tau_\pm)$ are at $k = 10$; Hopf points are invariant under $k \in \{5, 10, 20, 40\}$ at fixed δ — the sweep holds δ numerically fixed rather than maintaining the table relation $\delta = \log 2/k$, and this is the declared reading — and a sign change in ℓ_1 under that sweep would rearrange the lower window. The k-independence of equilibria and linearisations is a local statement only; fold locations depend on k, and no k-uniform topology is claimed.

5.3 Hopf persistence under residual feedback

Remark 5.1 (Local Hopf persistence, conditional). *Assume*

(H1) the five-state macro-reduction conjecture of the supplementary material — a slow-fast reduction whose fast-block uniform normal hyperbolicity is established as the proposition of S5, proved analytically on the parameter set $\alpha + \eta\beta < 1$, where α and β are the two coupling weights of the fast block's K–L Jacobian displayed in Supplementary S5, so that the inequality is that block's uniform-Hurwitz condition — with the sweep-quantified margin and the class-violation fraction recorded, and whose compact-memory-kernel condition for infinite-time persistence remains unverified,

(H2) the working-core projection of Section 2.4,

(H3) the working four-state (respectively three-state) characteristic equation has a simple pair $\pm i\omega$ at $\tau = \tau_\star \in \{\tau_-, \tau_+\}$ with $\mathrm{d}\,\mathrm{Re}\,\lambda/\mathrm{d}\tau \neq 0$ and no other imaginary eigenvalues,

(H4) the fast Jacobian remains uniformly Hurwitz at the joint equilibrium,

(H5) non-feedback mass compartments stay outside the delay loop.

Then: under the strict specialisation the core spectrum is a literal factor of the full characteristic function and the Hopf points persist exactly; under residual macroeconomic feedback of size ε the specialised system has a Hopf point $\tau_\star(\varepsilon) = \tau_\star + O(\varepsilon)$ in the ideal large-reservoir limit (add $O(1 - \sigma_{\mathrm{geo}})$ for the finite reservoir, σ_{geo} being the geological-reservoir adequacy ratio of Section 2.4).

The conditionality is mathematical content, and it is stated as such: the statement is conditional on the reduction's residual hypotheses — fast-block Hurwitzness beyond the verified parameter set of S5, and the memory-kernel condition — and the global fold events of Section 8 lie explicitly outside its hypotheses.

Derivation (conditional, in full). Under the strict specialisation, the block-triangular identity granted by the reduction conjecture factors the full characteristic function as $\Delta_{\text{full}}(\lambda,\tau) = \Delta_{\text{core}}(\lambda,\tau)\,\Delta_{\text{fast}}(\lambda,\tau)$ with Δ_{fast} uniformly Hurwitz on the declared set. The simple imaginary pair $\pm i\omega$ of the core is therefore a simple imaginary pair of the full function, and the implicit function theorem applied to $\Delta_{\text{full}}(\lambda,\tau)=0$ at $(\pm i\omega,\tau_\star)$ — where $\partial_\lambda\Delta_{\text{full}}\neq 0$ by simplicity — preserves the crossing exactly. Under residual macroeconomic feedback of size ε, write $\Delta_{\text{full}}(\lambda,\tau;\varepsilon) = \Delta_{\text{core}}(\lambda,\tau) + \varepsilon\,\chi(\lambda,\tau;\varepsilon)$ with χ continuous. Near $(\omega_\star,\tau_\star,0)$ the zero set of $F(\omega,\tau;\varepsilon) = (\operatorname{Re}\Delta_{\text{full}}(i\omega,\tau;\varepsilon),\,\operatorname{Im}\Delta_{\text{full}}(i\omega,\tau;\varepsilon))$ is a smooth curve through $(\omega_\star,\tau_\star)$ whenever $D_{(\omega,\tau)}F$ is nonsingular there. That nonsingularity is the Cauchy–Riemann restatement of the two hypotheses of the statement — simplicity of $\pm i\omega$ ($\partial_\lambda\Delta_{\text{core}}\neq 0$) and transversality ($\mathrm{d}\operatorname{Re}\lambda/\mathrm{d}\tau\neq 0$) — so the implicit function theorem yields $\tau_\star(\varepsilon)=\tau_\star+O(\varepsilon)$. The finite-reservoir $O(1-\sigma_{\text{geo}})$ term is the same argument with the perturbation scale replaced. The reduction conjecture itself remains the unproved hypothesis. □

5.4 The two-delay interpolation and the conditional mobilising-weight corollary

Linearising the interpolation (4) at the Candidate A point — an interior rest where both brackets vanish separately under the calibration of Section 2.5 — gives $\dot e = C_E e + C_m z(t-\tau_M) + C_p z(t-\tau_p)$, with $C_m = \chi_m C_Z^{\text{mob}}$, $C_p = \chi_p C_Z^{\text{prot}}$, and C_E the sum of the two gate-adjusted E-derivatives.

Proposition 5.1 (Two-delay characteristic identity). *The characteristic function of the stock–memory block coupled to the interpolated effort law is*

$$P(\lambda) - L(\lambda)\big(C_m e^{-\lambda\tau_M} + C_p e^{-\lambda\tau_p}\big) = 0, \tag{9}$$

with P and L the polynomials of Section 3.2.

Proof. The variational system of the interpolation has the same stock–memory block as (1) and the effort row $\dot e = C_E e + C_m z(t-\tau_M) + C_p z(t-\tau_p)$. Expanding the characteristic determinant along the effort row, the minor associated with $z(t-\tau_M)$ is exactly $L(\lambda)e^{-\lambda\tau_M}$ — the cofactor of the memory row, the same cofactor that produced (5) — and likewise for τ_p. □

Proposition 5.2 (Weighted small gain). *Assume*

(H1) the undelayed interpolation is stable (Hurwitz at $\tau_M=\tau_p=0$),

(H2) the zero root is excluded ($P(0)-L(0)(C_m+C_p)\neq 0$).

If

$$\sup_{\omega>0}\frac{(|C_m|+|C_p|)\,|L(i\omega)|}{|(i\omega-A_N)(i\omega+d)(i\omega-C_E)|} < 1, \tag{10}$$

then (9) has no purely imaginary root for any $\tau_M,\tau_p\geq 0$, and — since for retarded equations finitely many roots lie to the right of any vertical line and the right-half-plane root count changes only through imaginary-axis crossings — the equilibrium is exponentially stable for every fixed pair of delays (pointwise in the delays; the decay rate need not be delay-uniform).

Proof. An imaginary root $\lambda = i\omega$ with $\omega>0$ would satisfy $P(i\omega) = L(i\omega)(C_m e^{-i\omega\tau_M} + C_p e^{-i\omega\tau_p})$; taking moduli and using $|e^{-i\omega\tau}|=1$,

$$|P(i\omega)| \leq |L(i\omega)|(|C_m|+|C_p|),$$

contradicting (10), which is the triangle inequality reversed strictly. The excluded case $\omega=0$ is covered by the stated zero-root hypothesis; stability at the reference point is the stated undelayed-stability hypothesis; and the stability-switch principle (finitely many roots in the right half-plane, locally constant count off the imaginary axis; Hale and Verduyn Lunel, 1993, Ch. 11) transfers the reference stability to every delay pair. □

Corollary 5.1 (Mobilising weight, conditional). *At Candidate A the pure mobilising loop gain exceeds 1 and the pure protective loop gain is 0.080. Assume*

(H1) the common equilibrium and all linear coefficients depend continuously on χ_m,

(H2) the characteristic denominator remains nonzero on the imaginary axis,

(H3) the protective endpoint has a strict gain margin.

Then there exists a conservative threshold radius $\chi_m^* \in (0,1)$ *such that every interpolation with* $\chi_m < \chi_m^*$ *and* $\chi_p = 1 - \chi_m$ *satisfies (10): the small-gain exclusion remains valid on a nonzero protective-dominated weight interval.* ***Claimed:*** *existence of a sufficient-condition radius.* ***Not claimed:*** *identification of the actual onset weight for Hopf frequencies — failure of (10) does not imply Hopf, and* χ_m^* *is a sufficient-condition radius, not a physical critical weight. A Hopf of the interpolated system therefore requires a sufficiently large mobilising weight; it cannot be produced by decreasing* τ_p *alone.*

The corollary holds exactly under the listed hypotheses. The interpolation family is not otherwise established to preserve a common equilibrium or a nonvanishing denominator, and no promotion of the corollary beyond them is made.

Proof (in full). Define the certificate map

$$F(\chi_m) = \sup_{\omega>0} \frac{(|C_m| + |C_p|)\,|L(i\omega)|}{|(i\omega - A_N)(i\omega + d)(i\omega - C_E)|}, \qquad C_m = \chi_m C_Z^{\mathrm{mob}}, \quad C_p = (1-\chi_m) C_Z^{\mathrm{prot}}.$$

Hypothesis (H2) keeps the denominator nonzero on the imaginary axis. Because $|L(i\omega)|/|(i\omega - A_N)(i\omega + d)(i\omega - C_E)| = O(\omega^{-2})$ as $\omega \to \infty$, and because the coefficients A_N, d, C_E are bounded uniformly on $[0,1]$ (hypothesis (H1), continuity on a compact set), the tail $\omega > \overline{\omega}$ contributes less than $F(0)/2$ for some $\overline{\omega}$ uniformly in χ_m. The supremum is therefore a maximum over the fixed compact range $[\underline{\omega}, \overline{\omega}]$, on which the integrand is jointly continuous in (ω, χ_m), so F is continuous in χ_m (the maximum of a jointly continuous function over a fixed compact set is continuous). At the protective endpoint, $F(0)$ is the pure protective loop gain, $0.080 < 1$ (the stated value; hypothesis (H3) is its strict-margin form). Continuity supplies $\varepsilon > 0$ with $F(\chi_m) < 1$ for every $\chi_m < \varepsilon$; the conservative threshold radius is any $\chi_m^* \in (0, \min\{\varepsilon, 1\})$. For $\chi_m < \chi_m^*$ and $\chi_p = 1 - \chi_m$, (10) holds, which is Proposition 5.2's small-gain exclusion, valid for every pair (τ_M, τ_p). The final clause follows from the sufficient-condition status of the certificate: on the certified weight range no imaginary root can be produced by any τ_p, so a Hopf of the interpolated system requires $\chi_m \geq \chi_m^*$ — a sufficiently large mobilising weight — and cannot be produced by decreasing τ_p alone. □

For institutional design this says that, on the interpolated architecture, delay-induced cycles cannot be generated by accelerating the protective response alone; the mobilising weight must first cross a sufficient-condition threshold. This is a sufficient-condition statement, not an empirical calibration, and the actual onset of Hopf frequencies may lie below χ_m^*.

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6. The Protective Channel

The protective channel is the quota-tracking law of (3). Its behavioural reading for fisheries governance is direct: when an institution observes a deficit signal, effort is restored toward a cap rather than mobilised further; landing pressure is reduced as the signal worsens. The channel acts on the same stock–memory block as the mobilising channel, but with the sign of C_Z reversed and the gain modulus reorganised by the cap calibration. Its local mathematics is qualitatively different. There are no Hopf crossings at any delay at the calibrated point — a no-Hopf theorem — and the channel is stable under annual sample-and-hold review. The implication for institutional design is that, on this architecture, the calibrated quota-tracking law is the protective direction the loop should take, and the only sampled instability it exhibits is a discretisation artefact.

6.1 The quota-tracking law and its calibration

The protective law (3) with its calibration is stated in Section 2.5. One interpretation discipline travels with it. The effort variable may be an endogenous industry response, a legal quota-utilisation state, or an actual institutional control, and these interpretations are not interchangeable; the quota-tracking law is an institutional control law.

6.2 The no-Hopf theorem

Theorem 6.1 (Delay-independent local stability of the calibrated quota tracker). *Consider the Candidate A stock–memory linearisation together with the protective gains of Section 2.5. The statement is pointwise in the declared coefficients — not a theorem for every decreasing cap function or every protective law. The modulus cubic*

$$H(x) = x^3 + c_2x^2 + c_1x + c_0$$

has

$$c_2 = A_N^2 + d^2 + C_E^2 = 0.76339, \qquad c_1 = 0.028946, \qquad c_0 = 9.278 \times 10^{-6},$$

all positive, with $c_2c_1 - c_0 = 0.02209 > 0$. *Assume*
(H1) the undelayed Jacobian is Hurwitz,
(H2) the zero-root condition holds delay-independently, $P(0) \neq 0$.
Then the characteristic quasi-polynomial has no purely imaginary root for any delay $\tau_p \geq 0$, *and the equilibrium is exponentially stable for every* $\tau_p \geq 0$.

Proof. (a) *Exclude imaginary roots.* At $\lambda = i\omega$ the modulus balance of (5) reads $|P(i\omega)| = |C_Z||L(i\omega)|$; by Theorem 4.1 this is equivalent to $H(\omega^2) = 0$. For the protective channel, $C_E = -0.850336$ and $C_Z = -1.661702$ with the same $A_N = -0.0179$, $d = 0.2$, $B_E = 0.008955$, and the even-pairs cancellation gives

$$H(x) = (x + A_N^2)(x + d^2)(x + C_E^2) - C_Z^2B_E^2x = x^3 + c_2x^2 + c_1x + c_0$$

with $c_2 = A_N^2 + d^2 + C_E^2 = 0.76339$, $c_1 = A_N^2d^2 + A_N^2C_E^2 + d^2C_E^2 - C_Z^2B_E^2 = 0.028946$, $c_0 = A_N^2d^2C_E^2 = 9.278 \times 10^{-6}$. All three coefficients are strictly positive, so by Descartes' rule of signs $H(x) > 0$ for every $x \geq 0$ and no positive real root exists; no $\omega > 0$ solves the modulus balance. (The Routh–Hurwitz array of H, with $c_2 > 0$, $c_2c_1 - c_0 = 0.02209 > 0$, $c_0 > 0$, is the strictly stronger statement that all roots of H lie in the left half-plane; it is reported for completeness, while the undelayed Jacobian's own stability array is the separate polynomial of step (b) below. A symbolic sufficient condition for the no-root property is $c_1 = A_N^2d^2 + A_N^2C_E^2 + d^2C_E^2 - C_Z^2B_E^2 \geq 0$, the only coefficient of H that can change sign.)

(b) *Delay-independent stability.* The undelayed linearisation is Hurwitz: its characteristic polynomial is $P(\lambda) - C_ZL(\lambda)$ with $L(\lambda) = B_E\lambda$ (even-pairs cancellation), i.e. $\lambda^3 + 1.0682\,\lambda^2 + c_1'\lambda + c_0'$ with $c_1' = 0.2038$ and $c_0' = 0.00305$; all Routh conditions are satisfied (the protective $C_E = -0.850$ dominates, so the λ^2-coefficient $-A_N + d - C_E = 1.0682$). Explicitly, the Routh array for $P(\lambda) - C_ZB_E\lambda$ has first column entries $1,\ 1.0682,\ c,\ c_0'$, where $c = (1.0682\,c_1' - c_0')/1.0682$ is the standard third first-column entry of a cubic's Routh array (the quotient $(a_2a_1 - a_0)/a_2$ with $a_2 = 1.0682$, $a_1 = c_1'$, $a_0 = c_0'$), and both $c > 0$ and $c_0' > 0$. The zero root is excluded independently of delay: $P(0) - C_ZL(0) = P(0) \neq 0$ because $L(0) = 0$ and $P(0) = (-A_N)(d)(-C_E) = A_NdC_E \neq 0$. Since (a) shows the imaginary axis is never crossed for any $\tau_p \geq 0$, the stability-switch principle for retarded equations — finitely many characteristic roots lie to the right of any vertical line, and their count is locally constant while no root lies on the imaginary axis, so stability changes only through imaginary-axis crossings (Hale and Verduyn Lunel, 1993, Ch. 11) — transfers the

undelayed Hurwitz property of (b) to every fixed delay: the equilibrium is exponentially stable for each $\tau_p \geq 0$, pointwise in τ_p, without a delay-uniform decay rate being claimed. □

The channel-separation reading is the theorem's interpretation: destabilisation by short delay is confined to the mobilising summand, and the protective channel has no periodic branch born from the equilibrium at any delay.

The same conclusion is visible in the loop-gain form. Define

$$\Gamma(\omega) = \left| \frac{C_Z L(i\omega)}{(i\omega - A_N)(i\omega + d)(i\omega - C_E)} \right|.$$

Then Γ is continuous on $[0, \infty)$, $\Gamma(0) = 0$ (since $L(0) = 0$), $\Gamma(\omega) \to 0$ as $\omega \to \infty$ — linear over cubic, $O(\omega^{-2})$: the numerator carries $L(i\omega) = B_E i\omega$ — and a numerically located maximum gives $\sup_\omega \Gamma(\omega) = 0.08011 < 1$, attained at $\omega \approx 0.0589$ yr^{-1} (frequencies in yr^{-1}, the unit of C_E, d, A_N; Section 2). The loop-gain exclusion argument of Section 10 then excludes every imaginary root for all delays directly. The maximum is a numerical location, stated as such; the analytic certificate is the Descartes/Routh–Hurwitz argument above.

6.3 The iso-gain sign flip

Proposition 6.1 (Iso-gain sign flip). *Consider the gated Candidate A linearisation. Replacing C_Z by $-C_Z$, leaving every other coefficient unchanged:*

(i) leaves H identical (it depends on C_Z only through C_Z^2);

(ii) leaves the frequencies identical;

(iii) shifts each family's fundamental delay by π/ω on the branch that keeps it fundamental: the lower family ($\omega_1 \approx 0.02518$) moves up, $3.666 + \pi/\omega_1 = 128.374$ yr, and the upper family ($\omega_2 \approx 0.03944$) moves down, $150.358 - \pi/\omega_2 = 70.697$ yr, both remaining local Hopfs.

*The subscripts retain their original-family meaning — τ_- is the lower family's shifted delay, τ_+ the upper family's — so on the shifted axis the order is reversed: $\tau_+ < \tau_- < \tau_+^{\text{unshifted}}$. The reversed-gain linearisation has loop gain $1.016 > 1$ and retains the factor $\eta E^*/\Delta_{\text{ref}}$: it is **not** the quota law, whose genuine form changes the modulus as well as the sign. The proposition is the paper's sign-phase invariance statement: for a single delayed scalar feedback term $C_Z L(\lambda) e^{-\lambda\tau}$, replacing C_Z by $-C_Z$ leaves the imaginary-root frequency equation unchanged and shifts every admissible delay branch by an odd half-period — consequently, **feedback sign alone cannot create or eliminate Hopf-frequency families over the unrestricted delay axis**, and the protective no-Hopf property of Theorem 6.1 is a property of the quota law's modulus (its direct damping and delayed gain), not of its sign.*

This is the false-reversal identification hazard: a pure sign flip is not a protective institution, and attributing the reversed-gain crossings to a quota tracker would confuse two different effort laws.

Proof (in full). The imaginary-root condition of the gated Candidate A linearisation is $P(i\omega) - C_Z L(i\omega) e^{-i\omega\tau} = 0$ with the polynomials of (5). Separating modulus and phase,

$$|P(i\omega)| = |C_Z|\,|L(i\omega)|, \qquad \arg P(i\omega) - \arg\big(C_Z L(i\omega)\big) - \omega\tau \in 2\pi\mathbb{Z}.$$

The modulus equation depends on the effort coupling only through $|C_Z| = |-C_Z|$, so the frequency equation — the cubic in $x = \omega^2$ of Section 4 — and its certified positive roots are unchanged by the flip; the phase equation changes by exactly π, so every admissible delay branch is translated by an odd half-period, $(2k+1)\pi/\omega$. The certified fundamental pair and its frequencies are $(\tau_-, \tau_+) = (3.666149, 150.358477)$ yr and $(\omega_1, \omega_2) = (0.0251915, 0.0394366)$ (Section 5.1). The fundamental branch of each family is its smallest positive representative, which for the flipped gain is

$$\tau_-^{\mathrm{fl}} = \tau_- + \frac{\pi}{\omega_1} = 3.666149 + \frac{\pi}{0.0251915} = 128.374, \qquad \tau_+^{\mathrm{fl}} = \tau_+ - \frac{\pi}{\omega_2} = 150.358477 - \frac{\pi}{0.0394366} = 70.697,$$

the lower family moving up and the upper family moving down, as stated. Both shifted crossings remain simple and transverse — the certified evaluation gives $\mathrm{d}\,\mathrm{Re}\,\lambda/\mathrm{d}\tau = -6.07\times10^{-6}$ at $(\omega_1, 128.374)$ and $+1.83 \times 10^{-5}$ at $(\omega_2, 70.697)$, both nonzero — so each remains a local Hopf point of the flipped system. The invariance of the frequency equation is the stated sign-phase invariance: over the unrestricted delay axis the families are translated, never created or destroyed, and the protective no-Hopf property of Theorem 6.1 is accordingly a property of the quota law's modulus — its direct damping and delayed gain — not of its sign. □

6.4 Sampled protection and the discretisation crossing

Proposition 6.2 (Protective Euler-reviewed zero-order-hold monodromy). *Replace the protective delayed feedback by sample-and-hold of period T_r with one explicit Euler review step — an Euler-reviewed zero-order-hold controller, one particular sampled-data implementation among many (the standard reviewed controller of computer-controlled systems; Åström and Wittenmark, 1997), not sample-and-hold governance in general. Between reviews the variational system is $\dot{\xi} = A_{\mathrm{hold}}\xi$ with*

$$A_{\mathrm{hold}} = \begin{pmatrix} A_N & 0 & A_E \\ B_N & -d & B_E \\ 0 & 0 & 0 \end{pmatrix},$$

(the effort is frozen between reviews), and at each review — at the end of the interval, on the flowed state (the flow-then-update convention of Section 7) — the effort update is the explicit Euler step $e_{k+1} = e_k + T_r\big(C_E e_k + C_Z z(t_{k+1}^-)\big)$. The monodromy of one review interval is therefore

$$M_p(T_r) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & T_r C_Z & 1 + T_r C_E \end{pmatrix} \exp(A_{\mathrm{hold}} T_r). \tag{11}$$

Proof. On one interval $[kT_r, (k+1)T_r)$ the frozen-effort system is linear with matrix A_{hold} (the effort row is zero), so the state at the end of the interval is $\exp(A_{\mathrm{hold}}T_r)\xi_k$; the review then applies the affine update $\xi_{k+1} = \xi(t_{k+1}^-) + T_r(C_E e_{k+1}^- + C_Z z_{k+1}^-)\,\mathbf{e}_3$, i.e. the matrix factor displayed in (11). The sampled equilibrium is exponentially stable if and only if every eigenvalue of $M_p(T_r)$ lies in the open unit disc. □*

At the protective gains, $\rho(M_p(1)) = 0.9838 < 1$: annual review of the quota-tracking channel is linearly stable at Candidate A. On the grid $T_r \in [0.2, 20]$, the spectral radius is strictly below one on $[0.2, 2.306)$ and strictly above one on $(2.306, 20]$, so the Euler hold map crosses $\rho = 1$ at $T_r \approx 2.306$ (a grid resolution).

That crossing is carried by the discretisation, not by the governance system. Specifically, it comes from the interaction of the Euler-update factor $(1+T_rC_E)$ with the hold flow $\exp(A_{\mathrm{hold}}T_r)$ — the C_E row of the update restoring the coupling that the hold flow's zero third row lacks — and not from the scalar condition $1 + T_rC_E = 0$ alone, which (with $C_E = -0.850$) would place a threshold near $T_r \approx 1.176$, not at 2.306. So the crossing is a property of the Euler factor *combined with* the hold dynamics. It also sits within about 2% of the explicit-Euler stability limit of the scalar effort equation, $T_r < 2/|C_E| = 2.352$ yr (with $C_E = -0.850336$; the limit at which the scalar factor $1 + T_rC_E$ itself reaches -1), which makes the discretisation origin of the crossing self-evident. It is **not** a Hopf point of the protective delay equation, whose continuous spectrum Theorem 6.1 excludes at every delay. The finding is the mathematical statement: the $T_r = 2.306$ crossing belongs to the discretisation. The separating computation confirms it: under

the exact held-measurement update (Proposition 7.1) the protective monodromy has spectral radius below one at every tested review interval — 0.9838 at $T_r = 1$ yr and at most 0.9967 on $[0.2, 300]$ yr — so the crossing is carried entirely by the explicit-Euler command step, and the protective channel's sampled stability is not cadence-limited. The operator-specific scope is recorded: the mobilising hold map is unstable at $T_r = 1$ because the undelayed mobilising Jacobian is already unstable, and the protective map does not inherit that instability (Section 7).

6.5 Channel-specific pacing

Corollary 6.1 (Channel-specific pacing). *Three pacing statements hold.*

(i) For the mobilising bracket the equilibrium is linearly unstable for $0 < \tau < \tau_-$.

(ii) For the protective law at Candidate A the equilibrium is linearly stable for every $\tau_p \geq 0$.

(iii) For the two-channel system, in every connected parameter region where the undelayed interpolation is stable, the zero root is excluded, and the strict weighted small-gain bound (10) holds, stability is independent of both delays — any delay-induced stability loss must occur outside that certified region.

Proof. The first clause is Section 5.1 (undelayed instability with a stabilising lower crossing at τ_-). The second is Theorem 6.1. The third is Proposition 5.2 together with Corollary 5.1: wherever (10) holds, no imaginary root exists for any (τ_M, τ_p), so instability — if present — must originate in parameter regions where the mobilising weight is large, and is independent of the protective delay there. The synthesis inherits Corollary 5.1's interpolation hypotheses wherever its clause applies. □*

The policy-scope reading, instantiated: faster protective governance is not the hazard — the mobilising sign is.

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7. The Review Interval as Control

Periodic review is the form that institutional delay takes in modern fisheries governance: assessment is conducted on a fixed cadence (annual, biennial), and the assessment result is acted on with a discrete update. The review interval T_r — the **review cadence** — is the length of time between successive assessments, the interval at which the institution re-decides. On this architecture the review interval is a spectral design parameter that acts in opposite directions on the two channels. Periodic review is a *hybrid* system — a frozen-effort continuous hold interleaved with a discrete effort update at the review instants — and is not the delay equation of Sections 2–6 sampled at $\tau = T_r$. Its stability diagram is therefore not a re-plot of the continuous-delay τ-axis, and "lengthen the interval" is not a relabelling of "decrease the delay". The protective channel is stable at annual review under both the Euler and the exact update; its apparent 2.3-yr instability threshold is a discretisation artefact. The mobilising channel is unstable at annual review and restabilises as the review interval is lengthened — but its restabilising crossing is a complex unit-circle crossing of the sampled-data map, not a Hopf of the continuous-delay equation. The implication for institutional design is that the review interval is a controller knob, not a governance recommendation. An interval that stabilises the equilibrium may still allow severe inter-review depletion, and inter-review tube safety is an additional criterion not certified here. Throughout this section the *monodromy* is the one-period linear return map of the reviewed loop — the product of the held flow and the review update — whose spectral radius decides the linear stability of the sampled equilibrium (radius below one: stable; above one: unstable).

Theorem 7.1 (Sampled-data monodromy of the mobilising channel). *Linearise the gated three-state core about the interior equilibrium and replace the delayed feedback by sample-*

and-hold of period T_r with one Euler review step. Between reviews the variational system is $\dot{\xi} = A_{\text{hold}}\xi$ with A_{hold} as in Proposition 6.2. The monodromy of one review interval is

$$M(T_r) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & T_r C_Z & 1 + T_r C_E \end{pmatrix} \exp(A_{\text{hold}} T_r), \tag{12}$$

now with the mobilising gains $C_E = -0.0595$, $C_Z = +1.785$. Assume the flow-then-update timing convention (the pre-review state $\xi_k^- = \xi(t_k^-)$ flows for one period under the held effort, and the review applies the Euler update to the flowed state, giving the post-review state $\xi_{k+1}^- = M(T_r)\xi_k^-$ with M as displayed — the update factor acts on the end-of-period assessment; the reversed ordering $\exp(A_{\text{hold}}T_r)R(T_r)$ is a different hybrid system, not used here, the two orderings having closely related linear stabilities (proof); the nonlinear map differs, so the convention remains fixed throughout).

Then: (i) the sampled equilibrium is exponentially stable if and only if every eigenvalue of $M(T_r)$ lies in the open unit disc; (ii) a complex unit-circle crossing — the linear spectral signature of a Neimark–Sacker bifurcation (transverse crossing, nonresonance, and the nonlinear first Lyapunov coefficient of the map are not verified here, and the bifurcation name is reserved for that verified case) — occurs at those T_r where $M(T_r)$ has a simple pair $e^{\pm i\theta}$, $\theta \notin \{0, \pi\}$, with the remaining spectrum inside the disc; (iii) $(M(T_r) - I)/T_r$ converges to the continuous undelayed Jacobian as $T_r \to 0$. The statement concerns this sample-and-hold/Euler review scheme, not the continuous-delay equation.

Proof. The monodromy formula is Proposition 6.2's derivation with the mobilising gains; the stability criterion is the discrete-time spectral radius condition; the unit-circle characterisation is the standard eigenvalue-crossing criterion for maps. The consistency limit follows from $\exp(A_{\text{hold}}T_r) = I + A_{\text{hold}}T_r + O(T_r^2)$ and the update factor $I + T_r R_0$, where $R_0 = \mathbf{e}_3\big(C_Z\mathbf{e}_2^\top + C_E\mathbf{e}_3^\top\big)$ is the rank-one effort-update row, whose sum with $A_{\text{hold}}T_r$ is $I + T_r(A_{\text{hold}} + R_0) + O(T_r^2)$; writing $J_{\text{cont}} = A_{\text{hold}} + R_0$ for the full undelayed Jacobian of the continuous linearisation, the limit is $(M(T_r) - I)/T_r \to J_{\text{cont}}$ — R_0 is precisely what restores the C_E, C_Z entries that vanish in the third row of A_{hold}, which is why the consistency limit is not read off from A_{hold} alone. For the ordering comparison: $R_0\exp(A_{\text{hold}}T_r)$ and $\exp(A_{\text{hold}}T_r)R_0$ have identical spectra whenever R_0 is invertible (they are then similar matrices), so the linear stability classification transfers between the two orderings in that case; the nonlinear map differs, and the convention stays fixed. □

What Theorem 7.1 says. The theorem turns review-cadence stability into an eigenvalue test: compute the one-period return map $M(T_r)$ of the reviewed loop, and the sampled equilibrium is stable exactly when every eigenvalue of that map lies inside the unit disc. The rest of the section runs this test for the two channels.

On the gated Candidate A hold map the Euler scheme reports: annual review unstable, $\rho(M(1)) = 1.00055$ (the undelayed linearisation being already unstable, Section 5.1), with the sampled equilibrium's restabilising complex unit-circle crossing at $T_r^{\text{UC}} = 47.536$ yr and a multiplier reaching -1 at $T_r^{(-1)} = 79.143$ yr, and the reported stable interval $(47.536, 79.143)$ yr. These quantitative Euler values do **not** survive the exact held-measurement update; they are artefacts of the explicit-Euler command step and are recorded as such in Remark 7.1, with the exact record reported in Proposition 7.1 (unit-circle crossing at $T_r = 6.50$ yr; the Euler-reported crossings give exact spectral radius 0.786 at 47.536 yr and 0.597 at 79.143 yr). The load-bearing feature is therefore the *cadence* statement — lengthening the review interval restabilises a system that annual review destabilises — not the Euler map's particular thresholds. The computation is the zero set of $\det(M(T_r) - e^{i\theta}I) = 0$ on the declared map: for a slow stock under this review operator, **the review interval is a local spectral design parameter**.

Governance warning. Local spectral stability is not governance: a long review

interval may locally stabilise the equilibrium while allowing severe transient depletion between reviews, and inter-review tube safety — the stock floor held along the whole interval under the declared disturbance class — is the additional criterion a governance recommendation would require, which is not certified here.

The contrast with the protective channel — stable at annual review and at every tested interval under the exact update — is complete: the two sign channels respond to the review interval in opposite directions, and the protective channel's only sampled instability is the Euler command step's.

Remark 7.1 (Euler command-step artefact). *The Euler map's reported restabilisation at* $T_r^{\mathrm{UC}} = 47.536$ *yr with stable interval* $(47.536, 79.143)$ *yr is not a property of periodic review; Proposition 7.1 shows those crossings do not survive under the exact held-measurement update (exact spectral radius* 0.786 *at* 47.536 *yr and* 0.597 *at* 79.143 *yr, both below one) and that the exact monodromy has a single unit-circle crossing on* $[0.2, 300]$ *yr — the restabilising complex pair at* $T_r = 6.50$ *yr. The Euler thresholds are artefacts of the explicit command step, which is why they are labelled here and not reported as crossings of the review operator. The exact map and its single crossing are the quantities reported in Proposition 7.1.*

The design conclusion is not, however, an artefact of the update: as $T_r \to 0$ *every consistent scheme inherits the undelayed linearisation (Theorem 7.1(iii)), and as* $T_r \to \infty$ *the exact hold tends to a rank-one operator whose eigenvalue is the DC gain, proportional to* $L(0) = 0$ *by the filter identity, so the stock–effort coupling is lost and the equilibrium is stable for long enough review intervals. Each consistent scheme therefore restabilises at some* T_r*; only the location is scheme-dependent, quantified in the closing scheme-dependence remark of this section, which reports the tight* 6.50–6.73 *yr band across the three consistent schemes.*

Proposition 7.1 (Exact held-measurement monodromy). *Keep the sample-and-hold architecture, the frozen-effort hold flow* $\dot{\xi} = A_{\mathrm{hold}}\xi$*, and the flow-then-update timing convention of Proposition 6.2 and Theorem 7.1, and replace the explicit-Euler command step by the exact held-measurement update: the effort law* $\dot{e} = C_E e + C_Z z$ *is integrated exactly over the review interval with the measurement held at the flowed end-of-period value, giving the update*

$$e_{k+1} = e^{C_E T_r} e_k + \frac{e^{C_E T_r} - 1}{C_E}\, C_Z\, z(t_{k+1}^-).$$

The monodromy of one review interval is therefore

$$M_{\mathrm{ex}}(T_r) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & \dfrac{e^{C_E T_r} - 1}{C_E}\, C_Z & e^{C_E T_r} \end{pmatrix} \exp(A_{\mathrm{hold}} T_r), \tag{13}$$

which satisfies the same consistency limit as the Euler map, $(M_{\mathrm{ex}}(T_r) - I)/T_r \to J_{\mathrm{cont}}$ *as* $T_r \to 0$ *(the update factor is the exact exponential of the effort gain, of which* $1 + T_r C_E$ *is the first-order Taylor). Then, at the gated Candidate A point:*

(i) Protective channel ($C_E = -0.850336$*,* $C_Z = -1.661702$*).* $\rho(M_{\mathrm{ex}}(1)) = 0.9838$*, and* $\rho < 1$ *at every tested review interval on* $[0.2, 300]$ *yr, with maximum* 0.9967 *attained at the shortest interval tested. The protective hold map is stable at every review interval tested; the Euler crossing at* 2.306 *yr is carried entirely by the explicit-Euler command step and vanishes under the exact update.*

(ii) Mobilising channel ($C_E = -0.0595$*,* $C_Z = +1.785$*).* $\rho(M_{\mathrm{ex}}(1)) = 1.00035$*: the annual-review instability survives the exact update. The Euler-reported crossings do not: at* 47.536 *yr and* 79.143 *yr the exact spectral radius is* 0.786 *and* 0.597*. On* $[0.2, 300]$ *yr the exact monodromy has exactly one unit-circle crossing — the restabilising complex pair at*

$$T_r^{\mathrm{UC,ex}} = 6.50 \text{ yr},$$

with crossing eigenvalues $0.9846 \pm 0.1746\,i$ *(*$|\lambda| = 1$*), third eigenvalue* 0.1647 *inside the disc, a non-resonant argument (*$2\pi/\theta \approx 35.8$*), and transverse direction (*$\rho - 1$ *changes sign). The restabilisation is a property of review cadence, not of the command step. Holding the measurement at the start-of-period value instead of the flowed end-of-period value does not move the reported crossings at the reported precision. The Neimark–Sacker reading keeps the discipline of Theorem 7.1: the nonlinear conditions are not verified.*

Proof. The update is the exact solution of the scalar effort equation with z held constant; the monodromy factor follows from the flow-then-update convention, under which the update acts on the flowed state, whose effort coordinate is unchanged by the frozen hold. The consistency limit follows exactly as in Theorem 7.1 with $e^{C_E T_r} = 1 + C_E T_r + O(T_r^2)$ and $(e^{C_E T_r} - 1)/C_E = T_r + O(T_r^2)$. The channel statements are the computed spectral-radius records of M_{ex} on the declared grids, re-executable from the registered model constants; the same computation reproduces the registered Euler numbers of Proposition 6.2 and Theorem 7.1 before the exact map is evaluated, and both update conventions (start- and end-of-period measurement) agree at the reported precision. □

What Proposition 7.1 says. Under the exact update the protective channel stays inside the unit circle at every tested review interval, while the mobilising channel crosses it exactly once, restabilising at a review interval of about 6.5 yr; the Euler-reported half-century thresholds are artefacts of the command step, not of the cadence.

Remark (scheme-dependence of the sampled-data map). Neither M nor M_{ex} is the physically natural zero-order hold of the delay equation, in which the delayed signal z is held at its sampled value z_k while the effort dynamics $\dot{e} = C_E e + C_Z z_k$ runs continuously alongside the stock response. That scheme has monodromy, in closed form,

$$M_{\mathrm{ZOH}}(T_r) = e^{A_{\mathrm{Z}} T_r} + \Big(\int_0^{T_r} e^{A_{\mathrm{Z}} s}\, ds \Big) C_Z\, \mathbf{e}_3 \mathbf{e}_2^{\top},$$

with A_{Z} the full undelayed Jacobian with the C_Z entry removed. The two maps analysed above freeze effort throughout the flow and integrate the effort law impulsively at the review instant; they are one choice among several consistent schemes, so the phrases "the Euler-reported crossings are artefacts of the command step" and "the restabilisation is a property of review cadence" are properly read as scheme-dependent claims, not as properties of periodic review per se. The *qualitative* conclusion is not scheme-dependent, and it is provable without computing the location: as $T_r \to 0$ every consistent scheme inherits the undelayed linearisation ($(M - I)/T_r \to J_{\mathrm{cont}}$, Theorem 7.1(iii)); and as $T_r \to \infty$ the exact hold tends to a rank-one operator whose eigenvalue is the DC gain, which is proportional to $L(\lambda)$ evaluated at $\lambda = 0$ — and $L(0) = B_E \cdot 0 = 0$ by the filter identity of Section 3.2, so the map loses the stock–effort coupling and the equilibrium is stable for long enough review intervals. Hence every consistent periodic-review scheme restabilises at *some* T_r; only the location is scheme-dependent. The scheme-dependence is in fact small for this architecture. Re-executing the monodromy on the registered Candidate A constants: the exact held-measurement map has its mobilising complex crossing at $T_r = 6.5013$ yr; the same map with the measurement held at the *start*-of-period value gives $T_r = 6.5013$ yr (so the paper's clause that start-of-period measurement does not move the reported crossings holds); and the physically native zero-order hold M_{ZOH} gives $T_r = 6.7279$ yr (crossing eigenvalues $0.9855 \pm 0.1699\,i$), within about 4% of the exact value. All three consistent schemes therefore restabilise in a tight 6.50–6.73 yr band, and the reported 6.50 yr is a robust computed crossing of the declared map, not a certified invariant of periodic review. The protective channel is stable under M_{ZOH} as well (no crossing on $[0.2, 200]$ yr; $\rho(1) = 0.9928$), consistent with the protective no-Hopf claim. The rank-one limit is verified numerically: $\rho(M_{\mathrm{ZOH}})$ falls to 0.242, 0.0165, and below 10^{-6} at $T_r = 200$, 500, 2000 yr, with the map's numerical rank dropping to 2 at 500 yr and 1 at 2000 yr — confirming the DC-gain ($L(0) = 0$) loss of coupling.

The consolidated sampled-data record (channel × scheme × review interval). Every entry below is a record already stated in this section (Propositions 6.2 and 7.1, Theorem 7.1, and the scheme-dependence remark), collected in one table; the mobilising native-ZOH row's annual radius is not a recorded quantity of this paper.

Channel	Monodromy scheme	ρ at $T_r = 1$ yr	Unit-circle crossings on the tested grids
Protective	Euler (11)	0.9838	$\rho = 1$ at $T_r \approx 2.306$ yr (grid resolution; Euler command-step artefact, within 2% of the scalar Euler limit $2/\lvert C_E\rvert = 2.352$ yr)
Protective	Exact held-measurement (13)	0.9838	none on $[0.2, 300]$ yr; maximum 0.9967 at $T_r = 0.2$ yr
Protective	Native ZOH (M_{ZOH})	0.9928	none on $[0.2, 200]$ yr
Mobilising	Euler (12)	1.00055	complex pair at 47.536 yr and -1 at 79.143 yr (command-step artefacts; exact-map radii there 0.786 and 0.597)
Mobilising	Exact held-measurement (13)	1.00035	one complex pair at $T_r = 6.50$ yr ($0.9846 \pm 0.1746\,i$; third eigenvalue 0.1647)
Mobilising	Native ZOH (M_{ZOH})	not recorded	one complex pair at $T_r = 6.7279$ yr ($0.9855 \pm 0.1699\,i$)

Two readings travel with the table. First, the annual-review mobilising growth rate is small in absolute terms: $\rho - 1 = 5.5 \times 10^{-4}$ per review cycle (Euler) and 3.5×10^{-4} (exact) — e-folding times of roughly 1800 and 2900 review cycles respectively — so the annual-review instability is the $\tau = 0$ instability of Section 5.1 seen through a one-year hold that supplies too little phase, not a fast divergence. Second, the protective annual radius is the slow stock mode: the uncoupled modes of the protective hold-and-update map at annual cadence sit at $e^{A_N} \approx 0.982$ (stock; $A_N = -0.0179$), $e^{-d} \approx 0.819$ (memory; $d = 0.2$), and $1 + C_E \approx 0.150$ (effort; $C_E = -0.850336$), and the recorded $\rho(M_p(1)) = \rho(M_{\mathrm{ex}}(1)) = 0.9838$ — the same value under the two update schemes — is that stock mode, only weakly coupled by the controller. The protective maximum 0.9967 at $T_r = 0.2$ yr likewise matches the stock mode's $e^{A_N T_r} \approx 0.996$. The protective sampled-data record therefore says that the slow stock mode decays under review, not that the controller stabilises an otherwise unstable loop.

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8. Global Numerics at Declared Certification Levels

All results in this section are numerical results at their declared status: computed outputs tied to registered equations, parameter sets, history classes, methods, tolerances, and finite domains, with basins restricted to the histories actually tested. The fold-status discipline governs every global statement. The events below are numerical continuation, multiplier, basin, and turning-region results; the lower fold of Section 8.2 and the second fold of the same section are both certified at the discrete collocation level (the rebuilt Moore–Spence/Krawczyk stage for the

lower fold; the interval-Krawczyk certificate for the second fold — for each, the continuum off-grid residual stage remains unimplemented and the continuous-delay lift remains open); no continuous-delay fold proof is claimed. Rigorous saddle-node-of-periodic-orbit results for delay equations exist for specific classes (Beretka and Vas, 2020); none is claimed for the events below. The implication for fisheries governance is that, on the declared architecture, the global attractor topology is a five-regime structure whose lower and upper boundaries are both fold-certified at the discrete collocation level, and whose basin boundary lies between the two folds rather than at either; the saddle-node classification remains provisional.

8.1 Methods

Hopf points are located by root tracking of the characteristic equation and independently by the cubic (Theorem 4.1); folds by Keller pseudo-arclength continuation of the periodic orbit with 1.5–5 $\times 10^6$ yr persistence tests per step and far-from-equilibrium cross-checks (collocation-continuation numerics in the DDE-bifurcation tradition: Engelborghs, Luzyanina, and Roose, 2002; background: Guckenheimer and Holmes, 1983; Kuznetsov, 2004). Fixed-initial-condition bisection mislocates folds by more than 20 yr under the critical slowing down near each Hopf (linear rates 10^{-4}–10^{-5} yr^{-1}) and is not used.

8.2 The attractor topology and the lower boundary events

For the effort-bounded three-state core at Candidate A the lower boundary is carried by the pre-registered five-regime campaign (plan frozen before any run; executed once; pre-registered but not yet independently replicated; records deposited with the supplementary material, see Data availability). The record resolves the lower boundary into a **single fold at $\tau_f = 5.5872362$ yr of one S-shaped branch**: the small arm born at the subcritical Hopf $\tau_- = 3.666$ and the stable large arm are two arms of the same branch that meet at one fold. The fold is interval-certified at the discrete collocation level — the rebuilt Krawczyk stage of the A025 pipeline encloses the unique Moore–Spence zero of the $m = 64$ system in $[5.587236198689, 5.587236198691]$, the three-order solves ($m = 64/96/128$) agree to 2.69×10^{-11}, and both arms' dominant multipliers — real at every record, imaginary part identically zero to 10^{-6} — cross $+1$ at the fold: the campaign's recorded crossing proximity is 1.1×10^{-7} in τ, the fold-location certificate is the Krawczyk enclosure above, and the multiplier records are corroboration at their own, coarser, precision (small arm: 1.0192 at $\tau = 5.584$ falling to 0.9942 at 5.587; large arm: 0.2040 at $\tau = 4.0$ rising to 0.9692 at 5.5815 — τ recorded to three decimals, multipliers to four, so the tabulated points bracket the crossing without resolving it at the 10^{-7} proximity, which is the campaign's record, not an interpolation of the table). The two cycles coexisting at $\tau = 5.575$ are genuinely distinct (amplitudes 24.91 and 19.94; periods 322.60 and 308.16 yr) but are arms of one branch, not two families with two folds. The registered record places the basin-bisection endpoint 0.011 yr below the fold. The saddle-node-of-periodic-orbits classification remains **provisional** — the certification is of the discrete collocation system, the continuum off-grid residual stage is still open, and the crisis alternative survives as a reading of the archived record — and no Neimark–Sacker or torus event is involved in either reading (the multiplier is real throughout). The registered records reproduce the gate-floor facts near $\tau = 5.55$ ($E \leq 9.21$, $N \geq 68.6$): the termination is not a gate singularity, not a graze of $N = 0$, and not a homoclinic connection over the computed segment (the period is 324 yr and decreasing there; a homoclinic limit farther along the branch is not excluded by a finite segment).

The attractor sequence is the five-regime topology on the explored fundamental-delay range — $0 < \tau \lesssim 160$ yr, before the first recurrent branch of the lower frequency family (≈ 253 yr, Theorem 4.1), so the classification is a statement about the declared domain (Figure 1), not the entire positive delay axis: (i) $0 < \tau < \tau_-$, the equilibrium unstable with a single large-amplitude cycle the only attractor found from the declared tested histories. (ii) $\tau_- < \tau < \tau_f$,

a stable focus coexisting with the large cycle (lower bistable window, width ≈ 1.92 yr, closed at the fold $\tau_f = 5.5872362$ yr). (iii) $\tau_f < \tau < \tau_{f2}$, settling to the equilibrium from the tested histories (finite searches support, but cannot prove, basin monostability; the second-fold branch records show an unstable upper arm existing mathematically only above the second fold — over regime (iv)'s $64.4 < \tau < 150.4$ yr, as Section 8.3 registers — without generic basin). (iv) $\tau_{f2} < \tau < \tau_+$, mathematical bistability: the second S-branch's stable returning arm coexists with the still-stable equilibrium from the certified second fold $\tau_{f2} = 64.4023272$ yr, but the registered basin records find the stable arm generically unreachable near the fold — the capture onset in $[148.6, 149.5]$ yr is a basin boundary inside this window, not a fold. (v) $\tau > \tau_+$ within the explored range, the equilibrium unstable and the captured family the only attractor found from the declared tested histories. The registered basin archive resolves the lower-window asymmetry at $\tau = 5.575$: the depleted history (H2) is captured by the cycle while the large-stock (H1) and near-equilibrium (H3) histories settle. A basin diagnostic records that the H2 capture at $\tau = 5.6$ is a saddle-node ghost transient (the same run settles at four times the horizon), consistent with the collocation fold at 5.5872362 yr. In the ungated family (M3-U) the upper-window attractor is a period-1 limit cycle (at $\tau = 131.8$ yr ungated: Poincaré spread $< 5 \times 10^{-3}$, envelope constant to 0.005% over 2×10^6 yr, fundamental period ≈ 135.6 yr, all Floquet multipliers inside the unit circle with dominant nontrivial pair modulus ≈ 0.81; an independent method-of-steps RK45 reproduces the envelope). In the gated core the registered record's upper-window captured object is the $E \approx E_{\max}$ face cycle (below).

Regime	Delay range	Attractor record
(i)	$0 < \tau < \tau_-$	Equilibrium unstable; a single large-amplitude cycle the only attractor found from the declared tested histories.
(ii)	$\tau_- < \tau < \tau_f$	A stable focus coexisting with the large cycle — the lower bistable window, width ≈ 1.92 yr, closed at the fold $\tau_f = 5.5872362$ yr.
(iii)	$\tau_f < \tau < \tau_{f2}$	Settling to the equilibrium from the tested histories; finite searches support, but cannot prove, basin monostability.
(iv)	$\tau_{f2} < \tau < \tau_+$	Mathematical bistability: the second S-branch's stable returning arm coexists with the still-stable equilibrium; the capture onset in $[148.6, 149.5]$ yr is a basin boundary inside this window, not a fold.
(v)	$\tau > \tau_+$	Within the explored range, the equilibrium unstable and the captured family the only attractor found from the declared tested histories.

At the upper boundary the registered second-fold record (the pre-registered second-fold campaign; same deposit and status discipline as the five-regime campaign) resolves the boundary into a **certified second fold at** $\tau_{f2} \in [64.402327203368, 64.402327203372]$ **yr**: the upper branch, carried downward from the converged cycle at $\tau_+ - 0.05$, neither terminates at $\tau = 130$ nor fails collocation anywhere — 183 accepted natural steps at residuals $\leq 3.7 \times 10^{-12}$ from $\tau = 150.308$ to the turning-point stall at 64.402327895 — and turns at the interval-Krawczyk-

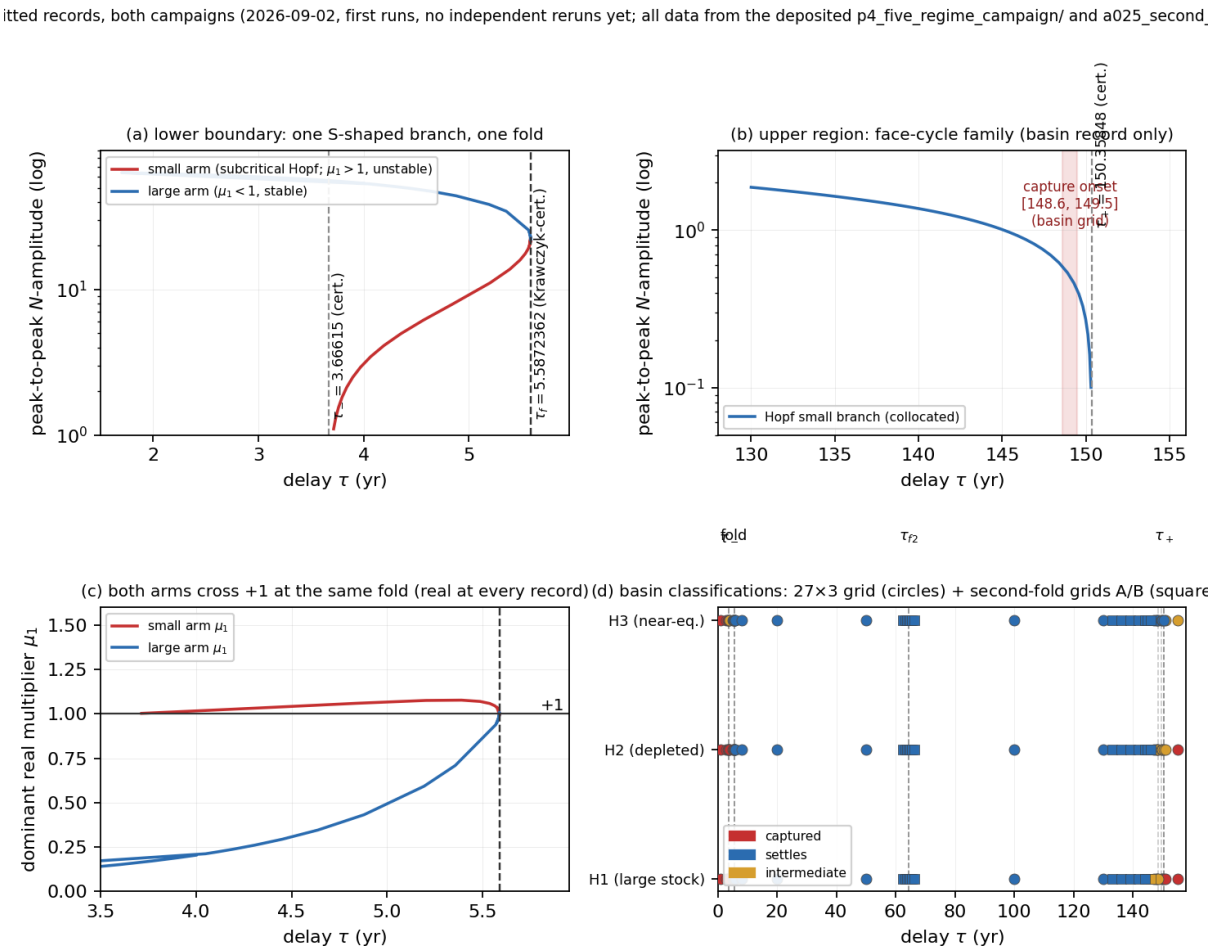


Figure 1: The five-regime attractor topology from the two pre-registered continuation records (five-regime campaign and second-fold campaign; every panel drawn from the deposited campaign CSVs, see Data availability). (a) The lower S-branch: the subcritical-Hopf small arm (red, $\mu_1 > 1$) and the stable large arm (blue, $\mu_1 < 1$) meet at the single Krawczyk-certified fold $\tau_f = 5.5872362$ yr. (b) The upper S-branch: the unstable arm from the Hopf at τ_+ folds at the certified $\tau_{f2} = 64.4023272$ yr and returns as a stable arm (inset: its real multiplier through the fold), with the basin capture onset $[148.6, 149.5]$ yr — a basin boundary, not a fold — and the certified $\tau_+ = 150.35848$ yr. (c) The lower branch's dominant real multipliers crossing +1 at the same fold. (d) Basin classifications (captured / settles / intermediate) on both registered grids (27×3 plus the second-fold grids A/B, dt $= 0.02$): the depleted history is captured at $\tau = 5.575$; every tested history settles from 5.6 to 146 (H1 intermediate at 145–146); capture returns only inside $[148.6, 149.5]$ yr.

certified Moore–Spence zero (unique in the box, G' nonsingular throughout, both nondegeneracy constants excluding zero; three-order agreement 5.53×10^{-7} at $m = 64/96/128$). The branch is a second S: the unstable subcritical arm from the Hopf at τ_+ (real multiplier > 1 at every record, peaking at 1.27745 at $\tau = 74.902$) meets the fold at amplitude ≈ 10.5, and sixteen pseudo-arclength points past the turn record a **stable** returning arm ($\mu_1 = 0.9993$ falling monotonically to 0.9435 at $\tau = 64.438$). **The stable arm is not generically reachable near the fold**: the campaign's basin grids find every tested history settling around τ_{f2} (21 runs at $\pm0.5/\pm1/\pm2$; dt-halving at 64.4 unchanged) and throughout the 133–146 gap (21 runs; H1 intermediate at 145–146, critical slowing toward the capture window). The capture onset $[148.6, 149.5]$ yr is therefore a **basin boundary** inside the mathematically bistable window (τ_{f2}, τ_+) — not the stable family's birth (that is τ_{f2}) and not the equilibrium's stability boundary (that is the certified τ_+). The captured object there remains the $E \approx E_{\max}$ face cycle (peak-to-peak $N \approx 44$; $E \in [0.35, 29.91]$ on the basin seeds), which no campaign could Fourier-collocate at $m = 64$ or $m = 128$ (two recorded collocation failures): the connection from the stable arm's last record ($\tau = 64.438$) to the face-cycle family is the natural conjecture, unverified over the 64.438–148.6 gap (a follow-up campaign would be required). The Hopf small branch is reproduced at both ends of the amplitude window (0.100 at $\tau = 150.31$ and 1.874 at $\tau = 130$). The face-riding cycle is reached from admissible histories in the upper window: the admissible topology of regimes (iv)–(v) is carried by the face cycle.

8.3 The registered numerical families

M3-U. Candidate A crossings 6.8814 / 132.3749 yr (Candidate B 6.2136 / 76.2906 yr); persistence boundaries near 7.355 and 131.24 yr; lower bistable window ≈ 0.47 yr, upper ≈ 1.1 yr; basin-dependent capture at $\tau = 131.8$ yr; the upper-window attractor a stable period-one cycle with all nontrivial computed Floquet multipliers inside the unit circle and the near-lower-Hopf small orbit unstable (dominant multiplier > 1). The saddle-node-of-periodic-orbits classification of either persistence boundary remains conjectural unless branch collision and nondegeneracy are demonstrated. Sensitivity sweeps over $\eta \in [0.5, 3.0]$ find zero or two positive cubic roots; the two-crossing window in r is bounded ($\approx (0.008, 0.022)$ yr^{-1} at $\eta = 0.914$, extending to ≈ 0.061 yr^{-1} at $\eta = 3$) — model sensitivity results, not empirical calibration. A Droop nutrient–quota variant of the same core — an external nutrient pool and an internal quota with quota-limited growth $\mu(q) = r(1 - q_{\min}/q)$, a growth-coupled pool, and the interior equilibrium preserved — leaves the r-window unchanged, with window upper edge ≤ 0.023 yr^{-1} at $\eta = 0.914$ and no Hopf crossing at any $r \geq 0.2$ yr^{-1}: a growth-coupled pool cannot be slow at large r (the quota self-relaxes at exactly r, and the nutrient pool's relaxation grows with the throughput; the only r-independent slow pool is the working core's ω_A-type exchange, which narrows the window).

M3-B. Candidate A crossings 3.67 / 150.36 yr (Candidate B 5.5128 / 80.4245 yr; no gated-Candidate-B global branch or fold result is registered). The lower global boundary is the single fold of one S-shaped branch at $\tau_f = 5.5872362$ yr (Krawczyk-certified at the discrete collocation level; both arms cross +1 there); the upper boundary is the certified second fold at $\tau_{f2} \in [64.402327203368, 64.402327203372]$ yr (interval Krawczyk on the $m = 64$ system; three-order agreement 5.53×10^{-7}), the upper branch an S whose stable returning arm is not generically reachable near the fold — the capture onset $[148.6, 149.5]$ yr (basin grid) is a basin boundary inside the bistable window (τ_{f2}, τ_+), and the fold type (saddle-node of periodic orbits) remains undetermined. Multiple far-from-equilibrium seeds failed to reach a large cycle throughout the tested interior range $5.57 < \tau < 148.3$ yr — finite searches that support, but cannot prove, interior monostability (basin-scoped: the second-fold branch records show an unstable upper arm existing mathematically on $64.4 < \tau < 150.4$ yr without generic basin, so the statement concerns basins, not branch existence). The inferred subcritical signatures (amplitude exponent 0.47; surrogate cubic coefficient $\approx 3.9\times10^{-6}$) are inferred numerical classifications, not centre-manifold Lyapunov coefficients; the two arms belong to one branch meeting at a single

fold, so a single square-root collision law applies in principle but is not verified (no continuum certificate); k-independence is local only. Ungated-Candidate-B global values are registered: the three-state ungated Candidate B has no lower fold (its lower crossing is supercritical, Section 5.2) and an upper fold of the large cycle at 76.075 yr against $\tau_+ = 76.2906$ yr — a narrow upper bistable window $(76.075, 76.29)$ — while the four-state ungated Candidate B carries the corresponding Hopf crossings at 6.25 and 76.33 yr with its upper fold not pinned; these are registered numerical values, their classification as saddle-nodes of periodic orbits remains open, and no gated-Candidate-B fold location follows from them.

M3-LC. Upper persistence boundary ≈ 132.0 yr at $\psi = 1$ versus ≈ 132.5 yr at $\psi = 0$ (raw outputs 131.998/132.499; separation 0.501 yr), with an inter-locator discrepancy of ≈ 0.8 yr against an independent locator's 131.24 yr taken as the localisation-uncertainty scale; the defensible conclusion is an order-of-one-year shift, not an exact percentage. At $\tau = 115$ yr, $\kappa = 0.5$, long-lived transients of order 10^4–10^5 yr reach minimum $N \approx 33$ under pure stock culling and $N \approx 10$ under pure recruitment suppression: the two channels of harvest pressure — removing standing stock versus suppressing recruitment — differ in the depth of their transient excursions, though the local bifurcation structure cannot see them. In the fixed-demand experiment ($D = 0.7 > S_{\max} = 0.5$, $N(0) = 50$), stock culling is stopped at its first hitting time of $N = 0$ (equivalently implemented with an explicit donor limiter; the unconstrained vector field is not continued into negative stock), placing pure culling at zero near time 158, while pure recruitment suppression approaches zero asymptotically and reaches $N < 1$ near time 430 — a first-hitting-time result. Local equivalence does not imply excursion equivalence; assigning ψ to a field system requires age-, stage-, or replenishment-specific evidence.

Four-state working core. Ungated Candidate A donor-limited equilibrium $(N^*, A^*, E^*) = (89.5256, 397.8665, 2.0896)$ with crossings 6.982022 / 132.272044 yr and persistence boundaries near 7.374 and 130.77 yr (upper bracketed $[130.770, 130.771]$); gated counterpart crossings ≈ 3.7849 / 150.12 yr with cycle periods $\approx$ 360–380 yr (lower regime) and 150–160 yr (upper regime). The turnover stability boundary at $\tau = 0$ is $\omega_A^* \approx 0.001316$ (the sweep's recorded value is 0.001316298; quoted at sweep resolution, not enclosure precision; gated ≈ 0.001330), located by a 1798-point equilibrium sweep, sixty sub-threshold simulations (ten ω_A values, six delays in $[0, 300]$ yr, all converging to the equilibrium), and sixty continuation points above it producing a continuous Hopf pair — finite sweeps supporting sub-threshold stability across the tested delays, not a theorem for all parameter values. The reduced open-pool caveat stands: freezing A is not a justified fast-variable elimination at the baseline $\omega_A = 10^{-3}$ yr^{-1}. The certified three-state second fold at 64.4023 yr falls inside the four-state's upper fold bracket $[64.25, 64.5]$ yr (persistent orbit of amplitude ≈ 11 at 64.5, continuation failure by 64.25): reproduced to within the bracket width — the bracket spans 0.25 yr, or 0.4% of the fold value, so the version-robustness of the upper fold location is a bracket-resolution statement — while the Hopf pair stays within a few percent and the basin reachability of the stable arm is version-specific; the characteristic-pinned detail is in Section 8.4.

MPF (primitive-flux core). At the illustrative baseline $(\mathcal{M}, \mu, K_A, d, c, \gamma_U, q, E_{\max}, \eta, \delta_0, \Delta_{\text{ref}}, \tau_m) = (100, 0.340, 24.5, 0.072, 0.00995, 0.388, 0.0384, 35.8, 2.23, 0.0118, 2.29, 5.13)$ the equilibrium is $(X^*, U^*, E^*) \approx (16.68, 10.23, 0.435)$, with no local Hopf crossing for $0 \le \tau \le 500$ (characteristic-root and argument-principle counts) and small perturbations decaying at all directly tested delays. The apparent onset near $\tau = 33.4$–33.6 is a long-lived decaying transient (return within $\approx 2 \times 10^4$ time units throughout the tested 33.4–34.8 interval); for $\tau \gtrsim 35$ a slow-fast oscillation persists for some tested memory histories and not others — basin-selective global dynamics, not a local Hopf or a classified periodic-orbit fold. The absence of a baseline Hopf is parametric: Hopf roots first appear at $\eta_{\text{crit}} \approx 2.337$; above τ_+ at $\eta = 10$ the attractor is diagnosed as irregular slow-fast intermittency (inter-excursion-interval coefficient of variation 1.58, return-map anticorrelation $r = -0.47$; the underlying global bifurcation mechanism is unresolved). The full η_{crit} sweep with the interleaving pair values and the pair-birth

structure, the intermittency diagnostics, and the sigmoid-gated effort screen (more than 300 parameterisations without finding a genuine delay-induced Hopf — a numerical negative result over the sampled domain, not a structural impossibility theorem) are deposited in the supplement (S11), with Section 10.4 carrying the screen's loop-gain reading. The MPF regime is neither the M3-B regime nor a transfer of its threshold values.

8.4 The four-state working core

The characteristic matrix $\Delta(\lambda) = \lambda I - A_0 - A_\tau e^{-\lambda\tau}$ of the gated working core at the frozen-donor equilibrium $(N, A, Z, E) = (89.52562, 397.8665, \log 2/10, 2.08962)$ has simple imaginary roots at

$$\tau_- = 3.78487 \text{ yr (period } 250.44 \text{ yr)}, \qquad \tau_+ = 150.12175 \text{ yr (period } 159.13 \text{ yr)},$$

with $|\det \Delta(i\omega, \tau)| < 10^{-18}$ — the characteristic-pinned pair, with τ_- 3.2% higher and τ_+ 0.2% lower than the three-state values — the directly computed spectral sensitivity of the crossings to the frozen-A perturbation, 12.7 times the parameter ratio $A_0/A^* \approx 2.5 \times 10^{-3}$, and not a consequence of the trajectory bound of Lemma 2.1, which bounds state deviations on $[0, T]$, not eigenvalue locations. Fourier collocation produces a small periodic orbit of residual $\sim 10^{-13}$ immediately below τ_+ (amplitude 0.090 at $\tau = 150.082$). Continuation of the large-amplitude cycle (converged orbit state and delay history carried between steps, 10^4–10^5 yr per step) locates the two global folds at

$$\tau^{(4)}_{\text{term},L} \approx 5.63 \text{ yr}, \qquad \tau^{(4)}_{\text{fold},R} \approx 64.4 \text{ yr},$$

the lower bracketed by a persistent periodic orbit at $\tau = 5.62$ (peak-to-peak N-amplitude ≈ 23) and failure of continuation by 5.64, the upper by a persistent periodic orbit at $\tau = 64.5$ (amplitude ≈ 11) and failure of continuation by 64.25. The resulting topology is a narrow lower bistable window $(3.78, 5.63)$ yr, a wide monostable interval $(5.63, 64.4)$ yr, and a wide upper bistable window $(64.4, 150.1)$ yr in which the large cycle coexists with the stable equilibrium but, from generic far-from-equilibrium histories, is reached only for τ above roughly 75–100 yr (depleted-stock histories only near $\tau \gtrsim 135$). Cycle periods run from ≈ 371 yr at $\tau = 4.5$ to ≈ 320 yr at the lower fold, and from ≈ 156 yr near τ_+ to ≈ 73 yr at the upper fold. Whether either fold is a saddle-node of periodic orbits is not established. A Neimark–Sacker bifurcation of a periodic orbit changes its stability and may create a torus but does not by itself terminate the branch, so branch termination, stability loss, and basin-capture loss are reported separately, and a torus-mediated reading would explain simulation-based loss of attraction without explaining branch nonexistence. A second bifurcation parameter invisible to the three-state core appears: at $\tau = 0$ the four-state equilibrium is unstable for $\omega_A > \omega_A^* \approx 0.001316$ and delay-independently stable below it; above it, on the ungated system, both Hopf points and both folds track smoothly, τ_- falling from ≈ 17.5 to ≈ 6.9 yr with the monostable interval remaining $\approx$ 120–260 yr; the gated $\tau_- \approx 3.78$ is the baseline (ω_A, κ_A) value, not a uniform constant. The coexistence of the characteristic-pinned pair with the sub-threshold delay-independent stability is not a contradiction, but the two objects must be labelled distinctly. The characteristic-pinned pair ($\tau_- = 3.78487$, $\tau_+ = 150.12175$ yr) is the spectrum of the frozen-donor characteristic matrix $\Delta(\lambda) = \lambda I - A_0 - A_\tau e^{-\lambda\tau}$ with the donor pool A held fixed at its equilibrium value — the standard quasi-polynomial spectrum of the reduced 4×4 system written with A pinned. The ω_A^* boundary (≈ 0.001316) is a different object: the dynamic-A turnover sweep at $\tau = 0$ of the full system (2), in which A is itself a state. Call the first the *frozen-donor characteristic pair* and the second the *dynamic-A stability boundary*; they are two different operators, and reporting both as "the four-state working core" invites exactly the confusion the distinction resolves. The frozen-donor pair is only a spectral statement about the quasi-polynomial with A frozen; it is not the spectrum of the dynamic-A system, whose $\tau = 0$ equilibrium is delay-independently

stable below ω_A^*. Oscillation periods throughout $\tau < \tau_-$ are 250–390 yr, essentially independent of τ: the frequency is pinned by r and τ_m, not by institutional delay, and uniformly raising r toward fish-like values destroys the oscillatory regime rather than compressing the period into a decade.

The frozen-donor quasi-equilibrium discipline governs all of these numbers: the working point requires continuing geological support (≈ 4.652 stock units per year — a companion-interface quantity, Remark of Section 2.4) and is not a rest point of the closed mass ledger; it is incompatible with the formal quasi-steady slaving target of Section 2.4; and the working-core thresholds are $\sigma_{\text{geo}} = 1$ properties.

8.5 Parameter windows

Within the parameter subregion where the Candidate A two-Hopf pair exists, one-at-a-time variation across the reported ranges preserves the pair and the lower-boundary structure. Outside the bounded r-window the system is delay-independently stable or lacks a positive equilibrium. τ_- lies in 3.7–7 yr across the two effort laws and both candidates, and in 4–25 yr across the full (r, η) rectangle; τ_+ is primarily controlled by the biological time $1/r$ but also depends on the effort-response chart — Candidates A and B share $r = 0.02$ and have $\tau_+ \approx$ 132–150 yr and $\approx$ 76–80 yr respectively across the two effort laws (A: 132.37 ungated, 150.36 gated; B: 76.29 ungated, 80.42 gated). Reported literature ranges of intrinsic growth rates are wider than the instability window; the quoted $r \in [0.005, 0.4]$ yr^{-1} range is not independently sourced in this paper, so the identification warning is scoped to the paper's own computed two-crossing window, $r \in (0.008, 0.06)$ yr^{-1} (Section 8.3) — against any plausible empirical spread of r, most assessed stocks fall outside the window. The dimensionless groups that fix N^*/K and $r\tau_\pm$ once effort is scaled by $E_{\max}$ — and the complementary fact that the separate effort scale is not identifiable from (N, Z) alone — are identification statements treated in the companion analyses of sampled governance and assessment architectures (under review).

8.6 The scaffold companion: registered and verified records

The three records that entered the archive as unreproducible "reproduction targets" are the completed, verified results of a distinct research object — the **scaffold** model of the companion flow-balance framework (a separate manuscript) — whose generating code and result files have been recovered and are reproduced here verbatim. The scaffold is **not** the model of Sections 2–6 and 9; it is a stage/age-structured Ricker-cascade core with a material block, capital stock, and gated effort (states X_A, X_J, P, U, A, K_C, E — seven implemented nonlinear states, of which the appendix prose calls the classical scaffold "nine states" and lists ten symbols), and its delay-feedback loop is the companion's, not (1). Status: these are verified, machine-precision-computed records, not open directions. The three records, each reproduced by the verbatim scripts:

(1) Elevated-forcing cod-class Hopf. The scaffold_item3_refine scan over the economic parameters (η, ζ, K_0, q) finds, for the cod-class life history ($g = 5$, $P_0 = 2$, $N_c = 60$, $d_A = 0.2$, $d_J = 0.6$), a genuine Hopf at $\tau^* = 43.29$ with period 263.4 time units (crossing period ≈ 263; $E^* = 0.245$, $\bar{X}_A^* = 54.2$, $g_0 = 1.000$) at the elevated-but-plausible economic setting $(\eta, \zeta, K_0, q) = (5.0, 0.8, 0.03, 0.01)$ — maxgap +0.1631. The reported values $\tau^* \approx 43$ and period ≈ 263. In the code symbol convention the savings parameter is ζ (the manuscript's ς), and the fuel-efficiency parameter is K_0; at $\eta = 5$, $\varsigma = 0.8$, $K_0 = 0.03$, $q = 0.01$ are confirmed.

(2) Life-history anchoring at default economics. The scaffold_item_3_anchor run over the three anchored classes gives delay-independent stability at the default economic settings: anchovy-class ($g = 1$, $\bar{X}_A^* \approx 27.7$), sprat-class ($g = 2$, ≈ 34.6), cod-class ($g = 5$, ≈ 54.8) all satisfy the interior survival condition with a negative modulus gap (-3×10^{-4} to -10^{-3}) and no crossing over the tested delay search — confirming no crossing at default economics. (P_0

values that violated the Step-2 survival condition produced no equilibrium — a self-consistency check.)

(3) Broad crossing search. The scaffold_hopf_search + items_1-2 scans over the biological parameters give: among the strong-candidate genuine Hopfs (periods ≈ 900–11,000 time units on the broad search, and ≈ 3,800–35,000 on the six classified crossings), every classified crossing has $\mathrm{d}\,\mathrm{Re}\,\lambda/\mathrm{d}\tau < 0$ (stabilising), with crossing rates 10^{-7}–10^{-9}, and **no clean two-crossing τ_-/τ_+ window** — the stable windows after the stabilising crossing are fragmented and narrow, so the scaffolding's delay-induced structure is stabilising-only, not the paper's two-crossing (τ_-, τ_+) organisation. This is a genuine structural difference between the scaffold and the paper's three-state core, and it is the section's actual content.

State-count flag. The scaffold's appendix prose states "nine states" but lists ten symbols (the mass-conserved block $(\bar{X}_A, X_J, P, U, A, G)$ plus V_N, Z, K_C, E); the implemented code carries seven nonlinear states (G fixed at its mass-balance value, V_N fixed at 0 at the interior equilibrium, Z folded into the characteristic equation via the $(1 + \tau_m\lambda)$ memory factor). The count should be stated as seven implemented nonlinear states (equivalently ten conceptual states, nine per the appendix label) in one place, not inconsistently across the prose.

Status. These are registered, verified numerical records of the scaffold companion, reproduced here from the recovered scripts; they are reproduced on the declared instances and carry no saddle-node or collocation-grade certificate. They are included because they delimit the scaffold's delay-induced instability structure, which differs from the paper's — a difference worth recording, not as open directions but as the verified companion record it is.

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9. The Delayed-Recruitment (Maturation-Delay) System

The two channels of Sections 5 and 6 act on the scalar gated core (1), whose recruitment is instantaneous logistic inflow. This section registers the applied analogue in which the same governance structure — a delayed feedback law carrying institutional memory, deployed effort, and a gated response — acts on a stock whose own delay is the maturation lag from spawning to recruitment. The computational machinery is the recovered stage-analysis code of the companion empirical study, archived verbatim and re-executed without modification (the recovery audit is deposited with the supplementary material); the records below reproduce the recorded stage-analysis results at every anchor, and the section is the registration counterpart of the reproduction targets of Section 8.6.

9.1 The registered system

The working system has the three states of the core — stock N, deficit memory Z, deployed effort E — with the stock compartment replaced by the Gurney–Blythe–Nisbet delayed-recruitment plant (the class already cited in the manuscript for its laboratory evidence): recruitment into the stock at regeneration rate r, at the cohort $N(t-g)$ of the spawning event one maturation delay g earlier,

$$\dot{N} = r\,N(t-g)\big(1 - N(t-g)/K\big) - q\,E\,N,$$

so the perceived deficit is extraction minus the currently maturing regeneration,

$$d = q\,E\,N - r\,N(t-g)\big(1 - N(t-g)/K\big),$$

and the filter and effort law are exactly those of (1),

$$\dot{Z} = \frac{1}{\tau_m}\big[\Phi_k(\ell) - Z\big],$$

$$\dot{E} = \left(1 - \frac{E}{E_{\max}}\right)\left[\eta\, E\Big(\frac{Z(t-\tau)}{\Delta_{\text{ref}}} - \frac{E}{E_{\max}}\Big) + \delta_0 \frac{Z(t-\tau)}{Z_{\text{ref}} + Z(t-\tau)}\right].$$

Here g is the **maturation delay** (the recruit-to-mature lag), entering the stock equation and the deficit signal; τ is the **institutional delay**, entering **only** the effort equation (as in the base core); and η is the **effort-response coefficient**. The multiplicative gate is $(1 - E/E_{\max})$, the same as in (1), and is *not* denoted g here (the recovered code and its results file fix these labels: g is the maturation delay, η the effort-response coefficient). The two switches are the section's analogue of the two channels of Sections 5 and 6: at $g = 0$ the maturation delay is switched off and the system reduces **exactly** to the base scalar core; at $\tau = 0$ the institutional delay is switched off (the institutional channel still acting, undelayed). Because at steady state $N(t - g) = N^*$, the equilibrium is identical to the base core's ($Z^* = \delta$, E^* from the same quadratic, $N^* = K(1 - qE^*/r)$), giving a clean comparison with the base windows.

9.2 Validation at zero maturation delay

At $g = 0$ (maturation delay off) the two-delay crossing criterion — the delayed-recruitment counterpart of the stability-switch analysis of Section 5.1 — returns the recorded base windows exactly: recruitment-rate windows $r \in (0.00796, 0.02191)$ at $\eta = 0.914$ and $r \in (0.00676, 0.06028)$ at $\eta = 3.0$, both reproduced by the verbatim re-run with zero deviation. These are the registered validation rows for the two-delay machinery: the delayed-recruitment crossing structure exists, at zero maturation lag, only on a bounded window of recruitment rates, and the window widens with the effort response η.

9.3 Fine-map institutional bands

With the institutional channel on, the $\tau = 0$-stable two-crossing structure occupies four recruitment-rate bands at $\eta = 0.914$, on the locus $r\,g \approx 1.5$–1.6: $g = 1$: $r \in (1.565, 1.585)$; $g = 2$: $r \in (0.77, 0.81)$; $g = 3$: $r \in (0.50, 0.55)$; $g = 5$: $r \in (0.28, 0.33)$. The re-run locates the same bands with refined edges (1.564–1.586, 0.767–0.805, 0.492–0.547, 0.260–0.332), within the recorded fine-map grid resolution of ± 0.03; the band cells carry two delay crossings each on the wide characteristic mesh (rightmost root ≈ -0.06). A longer maturation delay moves the band to lower recruitment rates on that product locus — $g = 7$ near $r \approx 0.17$, $g = 10$ near $r \approx 0.08$ — so the stronger and longer the maturation lag, the slower the stock for which it carries the same two-crossing structure. The band τ-windows at the band centres lie within documented governance lags ($g = 1$: $(1.6, 3.5)$ yr; $g = 2$: $(2.6, 7.8)$ yr; $g = 5$: $(9.9, 20.3)$ yr) with periods ~ 4, ~ 8, and ~ 17 yr respectively. The $g = 2$ window lies inside the documented 2–13 yr governance-lag distribution (Hocherman, Trop, and Ghermandi, 2025) with a decadal (~ 8 yr) period, which is the sharpest falsifiable empirical avenue: fast-maturing small pelagics under continuous harvest control rules.

A mesh-range caveat is registered with the fine map. The map's characteristic scan is a bounded mesh, and an isolated apparent $\tau = 0$-stable crossing cell at $g = 5$, $\eta = 3.0$, $r \approx 1.54$–1.60 fails on the wide mesh (true rightmost root $+0.244$, unstable — the cell is a cohort regime of the delayed plant); the recorded absence of institutional $\tau = 0$-stable cells at $\eta = 3.0$ is confirmed, and band cells are claimed stable only where the wide mesh confirms them. The bands and their widths are parameter-sensitive and are characterised at finite parameter resolution — no collocation-grade classification and no τ-converged Hopf-locus mapping is claimed.

9.4 Nonlinear ground truth

The verbatim nonlinear integrators reproduce the recorded cycle rows. The slow-stock class ($r = 0.02$, $g = 5$, $\tau = 0$) carries the registered 358.8-yr biological cohort cycle (re-run 358.7

yr; stock swinging $\approx$ 28.5–95.4, E saturating at $E_{\max}$), reinforcing the centuries-long-period untestability conclusion — the slow-r system is not even quietly stable once recruitment is lagged, so the institutional-delay question becomes moot at slow r. The fast class ($r = 0.5$, $g = 5$, $\tau = 0$) carries the $\sim$ 20-yr cohort cycle (re-run 20.1 yr). The institutional cycle at ($r = 0.3$, $g = 5$, $\tau = 10$) has recorded period 16.96 yr with amplitude 8.66 (re-run 16.95 yr; the amplitude is tail-window dependent, re-run 9.36 under the recovered integrators' tail convention), and at $\tau = 21$ the same configuration is stable — so the **institutional-delay** window in which the delayed-recruitment channel carries its cycle is bracketed ($\tau \approx 10$ carries the $\sim$ 17-yr cycle; $\tau \approx 21$ is stable), the stage analogue of the two families' fundamental-window structure of Section 5.1. The recorded $\eta = 3.0$ band-centre rows close the section: ($r = 1.57$, $g = 1$, $\tau = 2.5$) gives period 4.0 yr, amplitude 3.5 (recorded; re-run 4.00/3.49), and ($r = 0.8$, $g = 2$, $\tau = 5.5$) gives period 8.04 yr, amplitude 12.4 (recorded; re-run 8.05/12.35), both τ-converged.

9.5 The distinct two-stage companion

The recovered batch also contains a separately-provenanced object — the two-stage (adult/juvenile) harvest model with states X_A, X_J, Z, E, where the harvest channel is adult take versus juvenile take. That model is the subject of a separate companion article and is **not** the delayed-recruitment plant of this section: there g is the stage *duration* (a demographic residence timescale) and the only genuine delay is the institutional τ, whereas here g is the maturation *delay*. The two objects' crossing records must not be conflated. Its archived computation (adult-take Hopf at $\tau^* \approx 52.07$ yr, period 269.4 yr, on the $g = 5$ class; 16.76/121.80 yr on $g = 1$; 31.66/200.88 yr on $g = 2$; none for juvenile take) is reproduced byte-identical from the recovered `stage_core.py`, and is recorded in the supplementary material as that companion's table, not as a result of the present section.

9.6 Registration and status

The records of this section are certified by the registration campaign (its gate log is deposited with the supplementary material, S9; all gates pass): the $g = 0$ validation rows (G2), the fine-map bands with refined edges, the wide-mesh verification of the recorded cells and the mesh-range caveat (G3), and the nonlinear ground truth (G4). The fine-map table is deposited with the supplementary material (S9). The records are a reproduction on the recovered, declared instance — a continuous-time characteristic-root scan with RK4 verification (the DDE analogue of a multiplier scan) — not a proof for all parameters; no discrete sampled-review stage-map Floquet computation was ever built (the recovered readme states so explicitly), and no saddle-node, Neimark–Sacker, or collocation-grade classification is claimed for the stage records.

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10. The Loop-Gain Family

The two channels of Sections 5 and 6 look different on the stock–memory block, but they share a common linearised feedback identity. The unifying object is the loop-gain family: an ecological resolvent supplying admissible frequencies, an institutional gain that must close the loop, and a delay supplying phase. The gain condition is delay-independent; the phase condition rotates through match and past it. Non-monotone delay-stability is therefore possible and structurally natural in multi-frequency loop architectures. The implication for institutional design is that the two channels are two instances of one feedback identity, and the channel-separation theorem is a property of the institutional gain — its modulus and the phase of the direct damping — not of the sign alone.

10.1 The general feedback identity

The vector ledger and the reduced cores share a linearised feedback identity, not one nonlinear system. Writing the ledger in primitive fluxes — assimilation g, mortality m, decomposition d_U, harvest h — the signed depletion signal $\ell = h-(g-m) = -\dot{X}$ is an identity, and liquidation is $[\ell]_+$. Let $\boldsymbol{\xi}$ be the ecological perturbation, $\dot{\boldsymbol{\xi}} = \mathbf{J}\boldsymbol{\xi} + \mathbf{b}_E e$, with memory a gain-$\gamma_m$ filter of $-\mathbf{c}^\top\dot{\boldsymbol{\xi}}$. The linearised loop of every core in this paper is

$$\lambda - C_E - C_Z e^{-\lambda\tau}\,\frac{-\gamma_m\lambda\,\mathbf{c}^\top(\lambda\mathbf{I}-\mathbf{J})^{-1}\mathbf{b}_E}{1+\tau_m\lambda} = 0. \tag{14}$$

The three-state core is the 1×1 case ($\mathbf{J} = A_N$, $\mathbf{b}_E = A_E$, $\gamma_m = 1/2$); the primitive-flux (stoichiometric) core is the 2×2 stock–detritus block with $\gamma_m = 1$ (its signed memory $\ell = -\dot{X}$ is unregularised, which places it outside the $Z \geq 0$ invariance statement of Section 2.2). Both reductions are symbolic identities: the three systems share this characteristic identity but differ in their nonlinear objects — the upper fold is version-robust at bracket resolution (≈ 64.4 yr: certified three-state second fold 64.4023; four-state bracket $[64.25, 64.5]$, whose 0.4% width sets that resolution). The Hopf pair stays within a few percent, the equilibrium memory differs ($Z^* = \delta > 0$ versus 0), and the primitive-flux core has no baseline Hopf ($\eta_{\text{crit}} \approx 2.34$). The three ingredients are an ecological resolvent supplying admissible frequencies, an institutional gain $C_Z\gamma_m$ that must close the loop, and a delay supplying phase; the gain condition is τ-independent, the phase condition rotates through match and past it, and non-monotone delay-stability is possible and structurally natural in multi-frequency loop architectures (an open-set transversality argument for genericity is not supplied, and no generic claim is made).

10.2 The loop-gain exclusion theorem

Theorem 10.1 (Loop-gain exclusion of delay-induced Hopf). *Write the general feedback equation as $\lambda - C_E - C_Z e^{-\lambda\tau}G(\lambda) = 0$. Assume*

(H1) $\sup_{\omega\in\mathbb{R}} \frac{|C_Z|\,|G(i\omega)|}{|i\omega - C_E|} < 1$,

(H2) the undelayed system is Hurwitz,

(H3) the zero root is excluded for every delay ($-C_E - C_Z G(0) \neq 0$; in the three-state architecture $G(0) = 0$ and $C_E < 0$ supplies this).

Then there is no purely imaginary characteristic root for any $\tau \geq 0$, hence no delay-induced Hopf, and the equilibrium is exponentially stable for each fixed $\tau \geq 0$.

Proof. An imaginary root $\lambda = i\omega$ would give $i\omega - C_E = C_Z e^{-i\omega\tau}G(i\omega)$, hence $|i\omega - C_E| = |C_Z||G(i\omega)|$, contradicting the strict supremum (H1); the zero-root and reference-stability hypotheses (H2)–(H3) close the switch argument exactly as in Theorem 6.1. □

The companion Nyquist reading: a Hopf root exists if and only if the loop transfer $L(i\omega,\tau) = C_Z G(\lambda)e^{-\lambda\tau}/(\lambda - C_E)$ equals 1 for some $\omega > 0$, with the small-gain bound $|L| < 1$ as the uniform special case. A separate sufficient certificate in the same family is the Halanay-type delay-independent stability theorem, stated for systems of the form $\dot{x}(t) = A_0 x(t) + A_1 x(t-\tau)$ in a declared norm: if for some $\alpha_0 > \beta_0 \geq 0$ the logarithmic matrix measure satisfies $\mu_*(A_0) \leq -\alpha_0$ and $\|A_1\|_* \leq \beta_0$, then the zero solution is exponentially stable for every fixed $\tau \geq 0$, with decay rate the unique $\nu > 0$ solving $\nu = \alpha_0 - \beta_0 e^{\nu\tau}$ (Halanay, 1966; the decay rate is denoted ν here so that η remains the effort-response coefficient of Section 2 throughout). It is not a consequence of the scalar loop-gain formulation and is stated as its own theorem, not as "the" general certificate. Two qualifications travel with it: the decay rate depends on τ and tends to zero as the delay grows (the stability is pointwise in τ, not delay-uniform), and the condition is sufficient, not necessary — its failure removes the certificate without proving instability; in nonlinear applications the theorem is local after a declared linearisation unless a global incremental bound is proved. The named-system theorems of Sections 5.4 and 6.2 are instances of this certificate family at declared linearisations.

10.3 The logistic identification lemma

Lemma 10.1 (Logistic identification). *In the primitive-flux core take $A \gg K_A$ and set $\mu - d = r > 0$, $c = r/K$. Then, uniformly on a declared domain where X is bounded and A is bounded away from zero,*

$$g - m = rX\left(1 - \frac{X}{K}\right) + O\left(\mu X \frac{K_A}{A}\right),$$

with the remainder carrying flux units. The identification requires the mortality identification: $K_A \to 0$ without $\mu - d = r$ and $c = r/K$ yields $g \to \mu X$, which is not logistic.

Proof. Write $g - m = \mu X \frac{A}{K_A + A} - dX - cX^2$; for $A \gg K_A$, $\frac{A}{K_A+A} = 1 - \frac{K_A}{A} + O((K_A/A)^2)$, so $g - m = (\mu - d)X - cX^2 + \mu X\, O(K_A/A)$. With $\mu - d = r$ and $c = r/K$ this is $rX(1 - X/K) + O(K_A/A)$. Taking $K_A \to 0$ alone gives $\frac{A}{K_A+A} \to 1$, hence $g - m \to (\mu - d)X - cX^2$, which is μX only if $d = c = 0$; the logistic identification is a joint statement about mortality and saturation, not a limit of the assimilation flux alone. The claim concerns the linearised feedback identity, not the nonlinear objects, which also differ in the equilibrium memory level, the effort baseline, and the closed form of E^*. □

10.4 The saturating-gate negative screen

Replacing the autocatalytic factor $\eta E Z_\tau / \Delta_{\text{ref}}$ by a saturating gate $\sigma(Z_\tau/Z_0)$ whose linearisation at equilibrium decreases in deployed effort and contains no factor $\eta E^*/\Delta_{\text{ref}}$, a search of more than 300 randomised parameterisations found no genuine imaginary-axis root of the general feedback equation (14) — the screen record, with the method stack of Newton eigenvalue tracking, joint modulus minimisation, and nonlinear integration, and the nonexistence reading, is deposited in the supplement, S11, which also carries the MPF sweep and intermittency records. This is a numerical nonexistence report on a compact searched set, not an analytic exclusion on the parameter space, and it does not establish that autocatalysis is necessary for a Hopf pair — other non-autocatalytic feedback laws could supply sufficient loop gain and phase. The gated law (1) is not of this class — its C_Z contains both the gate factor and the autocatalytic factor, and it admits the Hopf pair. In the searched parameter set, removing the autocatalytic factor reduced the gain enough that no imaginary-axis root was detected.

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11. Discussion

11.1 The two channels as a design distinction

The sign separation is the paper's structural finding. On the identical stock–memory block the two rules carry opposite mathematics:

- **The mobilising law ($C_Z > 0$)** carries a subcritical Hopf pair and a five-regime attractor topology.
- **The protective quota-tracking law ($C_Z < 0$)** carries the no-Hopf theorem — stability at every delay — and stability under annual review.

The distinction is not an artefact of the calibration: the iso-gain sign flip (Proposition 6.1) shows that a sign flip alone, without the genuine quota law's modulus change, retains the Hopf pair at shifted delays and a loop gain above one — so the sign alone does not decide, and the protective channel's no-Hopf property is a property of the quota law's combined direct damping and delayed gain. On this architecture, institutions whose calibrated effort law combines weak direct damping with sufficient delayed mobilising gain are exposed to delay-induced cycles;

institutions tracking a quota with the calibrated damping are not. The channel-separation reading answers the question posed by the behavioural literature (Moxnes, 1998): the hazard is not slow governance as such, but slow governance of the calibrated mobilising law — weak direct damping with sufficient delayed gain, of which the sign is one ingredient among three (modulus, phase placement, and digital implementation).

11.2 The review interval as control

Under periodic review the two channels respond in opposite directions to the same instrument. The protective channel is stable at annual review and — under the exact held-measurement update (Proposition 7.1) — at every tested review interval. Its only sampled instability is the explicit-Euler command step's near $T_r \approx 2.3$ yr — an artefact to be recognised, not a design threshold; for the calibrated local continuous-delay model, protective delay does not generate a Hopf instability, which is the precise form of the claim, not a general immunity of quota-tracking institutions. The mobilising channel is unstable at annual review and restabilises as the review interval is lengthened, through its complex unit-circle crossing at $T_r = 6.50$ yr under the exact held-measurement update. The practical reading: for a slow stock governed by a mobilising effort law, frequent assessment and rapid response is destabilising, and within this operator the only computed spectral control is the review interval itself — a governance parameter, not an ecological one.

> **Management caution.** Local spectral stabilisation is not governance: an interval that stabilises the equilibrium may still admit severe inter-review depletion, so the statement stops at the spectral level (Section 7).

The result echoes, in a single-species institutional setting, the sampled-data principle that the sampling period is a controller design variable (Åström and Wittenmark, 1997), and it supplies the missing dynamical content to the observation that institutional timing structures resource governance (Ostrom, 1990). Where harvester forecasting can dampen the cycles induced by delayed knowledge of the harvested population state (Adamson and Hilker, 2020), the review interval is the design variable analysed here.

11.3 Generality: what carries beyond the analysed class

Three coordinates of the institutional loop are independent of the ecology they govern: the sign structure and modulus of the response law, the deployment delay, and the review cadence. The theorems of this paper fix what those coordinates do within the declared stock–memory–effort class — the delay window and its two certified crossings, the no-Hopf property of the calibrated quota law, the opposite cadence responses of the two rules — and the class itself is stock-agnostic: a harvested renewable stock under institutional control, of which fisheries are the motivating instance, with the liquidation channels of Section 1.1 (culling the standing stock, suppressing recruitment, decoupled consumption) marking how far the same loop reaches beyond a single-sector reading. What transfers to other periodically reviewed renewable-resource regimes is therefore the mechanism's form, not the numerical thresholds: the thresholds are theorems about the declared parameterisation, and any field application must re-identify the institutional coefficients — the gains, delays, and cadences of the governance loop — with the same standing as the biological ones.

For management readers the paper's technical vocabulary has a direct translation:

Technical term	Management reading
Mobilising law ($C_Z > 0$)	Effort-escalation rule: decline triggers more deployment

Technical term	Management reading
Protective quota-tracking law ($C_Z < 0$)	Catch/quota rule: harvest restored toward a cap
Hopf crossing	Onset or loss of oscillatory stock dynamics
Neimark–Sacker-type crossing	Quasi-periodic instability emerging under periodic review
Monodromy spectral radius	Stability margin of one review cycle
Review interval T_r	Assessment/management review cycle length
Sample-and-hold review	The harvest control rule in force between assessments — a total allowable catch (TAC) or effort decision set at each review and held fixed until the next
Subcritical Hopf crossing	A tipping point for cycles: the boundary where an oscillatory attractor is already present, so the cycle regime arrives with coexistence rather than a clean switch
Inter-assessment stability margin	The monodromy spectral radius in plain words — how far one review cycle's contraction of small deviations sits below one

Two cautions travel with the translation:

- **Discretisation artefacts are a general warning for management modelling, not a curiosity of this system.** In both channels an explicit first-order update manufactured thresholds (near 2.3 yr for the protective channel, the half-century restabilisation for the mobilising channel) that the exact update of the same reviewed loop does not carry, so a discrete-time management model can report stability boundaries that belong to its discretisation rather than to the institution.
- **Local spectral stabilisation is not governance.** An interval that stabilises the equilibrium may still admit severe inter-review depletion (Section 7), and the capture boundaries of Section 8 are basin statements, not thresholds a manager can read off a spectrum.

The design reading that survives both cautions is the paper's central one: the response rule's class is a design choice, the review interval is a control variable with opposite effects under the two rules, and the institutional coefficients are dynamical parameters to be identified, bounded, and designed against. The documented institutional timelines collected in the next subsection ground the scales of those coordinates in the management record.

11.4 Documented institutional timelines

The two timing coordinates of the analysed loop — the deployment delay and the review cadence — have documented values at management scale, and the northern cod (NAFO 2J3KL) record is the canonical instance. Assessment of that stock proceeded on an annual cycle while the estimated spawning-stock biomass fell from about 735 kt in the 1991 assessment to about 31 kt in the 1994 assessment (DFO, 2016); the directed-fishery moratorium was announced on 2 July 1992, and the renewed commercial fishery on 26 June 2024 — thirty-two years later (DFO, 2016, 2024). The collapse itself has been attributed to overexploitation under sustained harvest pressure (Hutchings and Myers, 1994), with the assessment-and-response chronology dissected as a management failure (Walters and Maguire, 1996).

For this paper the record does two things. It documents that institutional response delays of years, and management-cycle decisions spanning decades, are real objects at the scale the analysed windows occupy — the mobilising channel's delay window between crossings near 3.7 and 150 yr, and the stage-analysis bands of Section 9.3, whose $g = 2$ τ-window lies inside the documented 2–13 yr governance-lag distribution (Hocherman, Trop, and Ghermandi, 2025).

And it supplies the canonical instance of the inter-review depletion that the local spectral statements of Section 7 explicitly do not certify: a stock falling by a factor of more than twenty between annual assessments while the institutional response remained incremental.

The review cadence, likewise, is an actively managed design parameter in operating fisheries rather than a theoretical construct. Annual assessment and annual catch-setting are the historical norm for major stocks, and management agencies are actively weighing moves to multi-year assessment cycles; the assessment-frequency literature finds that lengthening the interval changes fishery performance — less frequent assessment reduced relative yield and increased the risk of stock depletion and interannual variation in yield in simulation (Li, Bence, and Brenden, 2016), with the same caution echoed for a large coastal shark fishery under multi-year assessment (Peterson et al., 2022). That literature treats the review interval as a management-design variable with risk consequences — the empirical counterpart of this paper's reading of T_r as a local spectral design parameter — and its direction of caution matches the theory's: the mobilising channel restabilises only for review intervals above about 6.5 yr, but local spectral stabilisation is not governance, and inter-review depletion is the uncertified cost that both the theorems and the assessment-frequency record place at the centre.

What the documented timelines ground is the scale of the two timing coordinates, not the coefficients of any theorem. No stock, effort law, or delay value in this paper is calibrated to a named fishery, and the northern cod record is a timescale grounding and a caution, not a calibration: the institutional coefficients — the gains, delays, and cadences of the governance loop — remain to be identified from field data before any threshold of the declared class is read against a fishery. A structured search across more than thirty resource systems for an unconfounded instance of the delayed institutional feedback mechanism — a responsive rule, an independently dateable lag, no dominant competing driver, and unit-level data — found no eligible case; that search and its null result are the subject of the companion sampled-governance paper, and they mark the empirical programme named in the conclusion as prospective. The documented record therefore supplies what a sustainability or management readership requires of a theory paper — real governance timelines at the analysed scales — without converting any theorem into a claim about a named institution.

11.5 Certification levels

The paper's results carry three distinct certification levels, and the distinction is maintained claim by claim:

1. **Proved theorems.** The cubic modulus condition, the even-pairs algebra, the no-Hopf theorem, the monodromy formulas, and the loop-gain exclusions.
2. **Interval certificates.** The Hopf delay enclosures are interval-Newton certificates of the local spectrum of the cubic — re-execution-verified with outward rounding — but are not certificates of the bifurcation in the sense of Church and Lessard (2022); upgrading them to that standard is a stated open task. The 13-digit enclosures certify roots for the exact registered decimal parameter vector; they do not represent parameter-estimation uncertainty, and any policy reading of the thresholds must add a sensitivity layer over the declared parameter windows (Section 8.5).
3. **Declared-status numerical results.** The global folds, the attractor classification, and the Lyapunov coefficients are numerical results at declared status, with the lower boundary events left as a provisional saddle-node against a crisis reading, and the fold certification registered for both folds at the discrete collocation level (the lower fold's rebuilt Moore–Spence/Krawczyk stage, and the second fold's interval-Krawczyk certificate on the $m = 64$ system with three-order agreement 5.53×10^{-7}; the continuum off-grid residual stage and the continuous-delay lift remain open for both), documented component by component in the supplementary material.

Rigorous saddle-node results for delay equations exist for specific classes (Beretka and Vas, 2020); closing the gap for this system is a second stated open task.

11.6 Relation to the early-warning literature

The regime-shift literature's early-warning indicators — rising autocorrelation, rising variance — are candidate diagnostics near fold-type transitions under additive noise with responsive control (Scheffer and Carpenter, 2003; Scheffer et al., 2009; Carpenter et al., 2011). Four indicators are candidate diagnostics for the fold events of Section 8: critical slowing down (the lag-1 autocorrelation of the stock rises near a fold); rising variance (rolling variance increases under additive noise near a bifurcation); epistemic divergence (disagreement among independent estimators, innovation residuals, or a widening set-membership assessment tube — the true state is not observed, so the raw gap $|\hat{N} - N|$ between an estimated and the true stock (written in the paper's stock symbol, since $S(N)$ is the yield) is not itself an operational statistic, and the observable substitutes are listed instead); and policy inertia (physical stress increasing without a corresponding control adjustment). The caveat is load-bearing: these are not universal early-warning signals — rising autocorrelation and variance are tied to specific local bifurcations under specific noise and timescale assumptions, while the capture onset of Section 8.2 is a basin boundary — the certified second fold sits 84 yr lower in τ — and the lower fold's crisis alternative remains live, so an indicator tuned to a fold threshold would be watching the wrong event, and each indicator must be tied to a hypothesised mechanism before it is used; they do not follow from the local machinery of this paper, and each is a separate falsifiable empirical-statistical claim. In the bistable windows of Section 8.2 the relevant diagnostic is not a pre-collapse trend but the history-dependence of capture — the registered basin record's reversal at $\tau = 5.575$ (the depleted history captured, the large-stock history settling) is the load-bearing asymmetry — which the variance/autocorrelation pair cannot see.

11.7 The ecological reading

(i) **The exposed life histories are doubly selected.** With the maturation delay switched on, the two-crossing structure survives only on the product locus $r\,g \approx 1.5$–1.6 (Section 9.3): longer maturation delays move the band to slower stocks, down to $g = 10$ near $r \approx 0.08$. Selection acts a second time through the institutional delay. The band τ-windows grow with the maturation lag — $g = 1$: $(1.6, 3.5)$ yr; $g = 2$: $(2.6, 7.8)$ yr; $g = 5$: $(9.9, 20.3)$ yr — and the $g = 2$ window is the one that lies inside the documented 2–13 yr governance-lag distribution (Hocherman, Trop, and Ghermandi, 2025). Fast-maturing small pelagics are therefore the predicted exposed class, not an illustrative example. At the slow end the question inverts: with recruitment lagged, the slow-r system carries a 358.8-yr cohort cycle, so the institutional-delay question becomes moot there (Section 9.4).

(ii) **Ecological delay and institutional delay sit in different compartments and produce different cycles.** The maturation delay enters the stock equation and the deficit signal; the institutional delay enters only the effort equation (Section 9.1). Their cycles differ accordingly: cohort cycles whose periods track the life history (~ 20 yr at $r = 0.5$, $g = 5$; 358.8 yr at $r = 0.02$, $g = 5$) versus loop cycles whose period tracks the deployed response (16.96 yr at $r = 0.3$, $g = 5$, $\tau = 10$). With both delays present the institutional window survives on top of the ecological one — $\tau \approx 10$ carries the ~ 17-yr cycle; $\tau \approx 21$ is stable — the stage analogue of the fundamental window of Section 5.1 (Section 9.4).

(iii) **The form of extraction is ecological even where the local mathematics is not.** The two harvest channels of M3-LC — removing standing stock versus suppressing recruitment — leave the equilibrium, the Jacobian, the characteristic equation, and both

Hopf points identical (Section 2.3). The collapse dynamics still separate them: at $\tau = 115$ yr, $\kappa = 0.5$, long-lived transients reach minimum $N \approx 33$ under pure stock culling and $N \approx 10$ under pure recruitment suppression; under fixed demand, culling first hits $N = 0$ near time 158 while suppression approaches zero asymptotically, reaching $N < 1$ near time 430 (Section 8.3). Local spectra cannot see the extraction form; the excursion record can.

(iv) **Growth-coupled ecology cannot widen the window.** The Droop nutrient–quota variant leaves the r-window unchanged — upper edge ≤ 0.023 yr^{-1} at $\eta = 0.914$, and no Hopf crossing at any $r \geq 0.2$ yr^{-1} — because a growth-coupled pool self-relaxes at exactly r and cannot supply the slow companion variable (Section 8.3). The only r-independent slow pool in the registered family is the working core's ω_A-type exchange: decoupled storage, not growth-coupled ecology, is what can slow the loop.

Each reading consolidates records registered in Sections 2.3, 9, and 8.3; none adds a computation, an archetype, or a calibration, and the limitations of Section 11.8 apply to all four unchanged.

11.8 Limitations

(i) The model class is a mathematical parameterisation: no stock, effort law, or delay value is calibrated to a named fishery, and the institutional coefficients have not been identified from field data; the results are theorems about the declared class, with the parameter windows of Section 8.5 delimiting the regime of the two-crossing structure. The documented governance timelines of Section 11.4 ground the scales of the delay and review coordinates; they are not a calibration, and no institutional coefficient is identified from them. Identification is its own layer: different gain–delay–memory triples can realise similar observed harvest trajectories, so the institutional coordinates are latent structural parameters — separable only through exogenous variation or direct process measurement, not from the observed trajectory alone — and the functional form of the response law is identified, if at all, separately from its parameters. (ii) The delay is a single discrete lag — a reduced compression of the observation, assessment, decision, deployment, and compliance stages that a multi-stage institution carries separately; the documented cod chronology of Section 11.4 (the annual assessment cycle, the 1992 moratorium decision, the 2024 reopening) is a record of such stages at management scale, and what the single lag absorbs is their composite timing, not any one stage. Distributed delays and variable-time institutional lags require separate analysis. (iii) The sample-and-hold results concern the declared Euler review scheme and its exact held-measurement counterpart (Theorem 7.1, Proposition 7.1); other discretisations have different monodromies, and the continuous-delay and periodic-review recommendations are not interchangeable. (iv) The global classification of Section 8 is provisional where stated: the lower boundary's saddle-node classification, the upper boundary's type, and interior monostability are open. (v) The Hopf-persistence statements are conditional on the unproved reduction conjecture (Section 5.3). (vi) The scalar core of Sections 2–6 and the global numerics of Section 8 concern a scalar-logistic ecology; stage structure enters in the delayed-recruitment (maturation-delay) generalisation of Section 9, and spatial structure, multispecies interactions, and the distinct two-stage adult/juvenile harvest model (the separate companion of Section 9.5) remain outside the analysed class, with the delayed-recruitment literature (Gurney, Blythe, and Nisbet, 1980; Costantino et al., 1995) engaged as background.

11.9 Open problems

Two genuine next theorems are stated with their declared gaps, with the first Lyapunov coefficient of the exact-hold map's Neimark–Sacker crossing as the matching open computation: the Church–Lessard upgrade of the Hopf certificates (Section 11.5's first stated open task, taken together with the continuum off-grid residual stage of both folds), and persistence of a transverse fold of periodic orbits under small typed coupling (requires a verified fold baseline, spectral separation, and regularity of the infinite-dimensional Poincaré map). The further programme directions carried by the open-problem register — an RFDE/hybrid transition-persistence analogue, an n-patch super-equilibrium criterion, a variable-time delayed-hybrid information kernel, a restricted delay-separation principle for modularly identified governance loops, and an exergy-limited controller class with its declared conjecture and non-universality caveat — are recorded in the supplement's open-problem register (S6) and are not elaborated here: none connects to a theorem proved in this paper. A third stated open task is parameter-box certification: the interval enclosures of Section 5.1 certify the registered decimal parameter vector, and the one-at-a-time windows of Section 8.5 are not boxes — extending the interval Newton stage from the point vector to declared parameter boxes is the upgrade that would carry the threshold statements from arithmetic precision to the policy robustness that Section 11.5 reserves to a sensitivity layer.

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12. Conclusion

Where the delay sits determines what it does. Delays in the ecological dynamics of harvested stocks are the classical subject of delayed-logistic and delayed-recruitment analysis. The delay analysed here sits in the institutional loop that converts observed decline into deployed effort, and its mathematics separates sharply across the two calibrated response laws.

The mobilising channel — effort responds to decline by deploying more effort — carries a subcritical Hopf pair with interval-certified fundamental crossings of two positive frequency families, and a five-regime attractor topology on the explored fundamental-delay range whose global boundaries are two certified folds at the discrete collocation level: the lower S-branch fold and the second fold of Section 8.2, with the 148.6–149.5 capture onset being a basin boundary rather than a fold (for each fold the continuum off-grid residual stage and the continuous-delay lift remain open, and the saddle-node classification is provisional). Under periodic-review governance, the mobilising channel is unstable at the annual scale and restabilises only for review intervals above about 6.5 yr, via a Neimark–Sacker-type crossing of the exact held-measurement monodromy; the half-century crossing (47.5 yr) reported by a first-order (Euler) discretisation is a numerical artefact — a local spectral statement, not a governance recommendation. The protective channel — effort is restored toward a quota — carries a no-Hopf theorem valid at every delay at the calibrated point, stability under annual review, and an instability threshold that is provably an artefact of the discretisation.

The unifying object is the loop-gain family: an ecological resolvent, an institutional gain, and a delay supplying phase, with the gain condition delay-independent and the phase condition rotating through match and past it. For the design of extractive governance the conclusion is specific. The laws differ not merely in sign but in modulus, damping, phase placement, and digital implementation. The calibrated mobilising law is the hazard on this architecture. The review interval is a control variable with opposite effects in the two channels. And the institutional coefficients — the gains and delays of the governance loop — are dynamical parameters with the same standing as the biological ones, to be identified, bounded, and designed against.

Beyond the declared class, these coordinates provide a modelling vocabulary only for institutions that can be represented by the same stock–memory–effort structure. This does not establish empirical transfer: the theorems fix local effects for the analysed parameterisations, while field applications require assessing model structure and identifying the relevant gains, delays, and review cadence. The empirical programme is to conduct that identification with field data.

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Supplementary material

The full interval-enclosure table (enclosures for both candidates and both gating variants, with the verified root intervals in $x = \omega^2$ and the branch-safe phase evaluations), the collocation formulation details (mesh $m = 64/96/128$ and the small-branch computation), the fold-certificate gap documentation (the open continuum routes — the free-τ enclosure, the transversality and curvature conditions, and the continuous-delay lift — component by component; S3's status list describes the pre-rebuild state of the A025 fold computation; the status note, S12, records the current state), the variant registry (the response-sign hypotheses and the registered variants with their obligations), the statement inventory, the delayed-recruitment registration records

(S9), and the expanded proofs of Corollary 5.1 and Proposition 6.1 (S10, which labels these two objects Corollary 3 and Proposition 5; the label mapping is given in S12) are provided in the accompanying file `paper4_supplementary_v8.md` (S1–S10), together with the MPF material of S11 (the η_{crit} sweep and pair-birth structure, the slow-fast intermittency diagnostics, and the sigmoid-gated effort screen record).

Declarations

Data availability

All computations are deterministic and re-executable from the registered model registry and runners; the interval pipeline reproduces the displayed enclosures exactly with outward rounding. The five-regime continuation records of Section 8.2 (branch, Floquet, basin, solver, and environment archives; 148 branch records and 87 basin runs) are deposited at `research_program/validated_computations/p4_five_regime_campaign/` with their pre-registered plan and execution report; the rebuilt fold-certification stage sits beside the A025 pipeline at `research_program/validated_computations/a025_fold/`; the second-fold records of Section 8.2 (200 branch records, 45 basin runs, the interval-Krawczyk certificate and its preserved frozen-box failure record) are deposited at `research_program/validated_computations/a025_second_fold/` with their pre-registered plan and execution report. The parameter tables, the certified enclosures, and the numerical protocols are available from the authors and are deposited with the supplementary material. The recovered stage-analysis machinery of Section 9 is archived verbatim with the deposited material and re-executable; its recovery audit and the 21-gate registration campaign that certifies the Section 9 records (gate log and fine-map table, Supplementary S9) are deposited with the supplementary material.

Declaration of competing interest

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